

TECHNOLOGY



Microsoft Certified Azure Administrator: AZ-104

Monitor and Back Up Azure Resources



A Day in the Life of an Azure Architect

A company stores critical compliance information on Azure file shares. An Azure Administrator must ensure this content can be recovered if there's data loss or corruption. They must configure backup and restore policies that meet the company's regulatory needs.

The company has several critical virtual machine workloads running on Azure. An Azure Administrator must ensure the company can recover these virtual machines if there's data loss or corruption.

A vital component of production applications is logging and monitoring the health of the services. Azure Administrators determine the causes of failures and try to identify any problems before they occur.



Learning Objectives

By the end of this lesson, you will be able to:

- Configure Azure backup to back up and restore the data in the Microsoft cloud
- Configure Azure resource monitoring for a comprehensive solution for collecting, analyzing, and performing on data
- Analyze logs and metrics to gather all the data by Azure monitor
- List network watcher to monitor, diagnose, and view metrics, and enable or disable logs for resources in an Azure virtual network



Administer Data Protection

Azure Backup

It is the Azure-based service a user can use to back up (or protect) and restore the data in the Microsoft cloud.

Benefits

- Automatic storage management
- Multiple storage options
- Unlimited data transfer
- Data encryption
- Application-consistent backup
- Long-term retention



Implement Azure Backup Center

Backup center provides a unified management experience in Azure for enterprises to govern, monitor, operate, and analyze backups at scale.

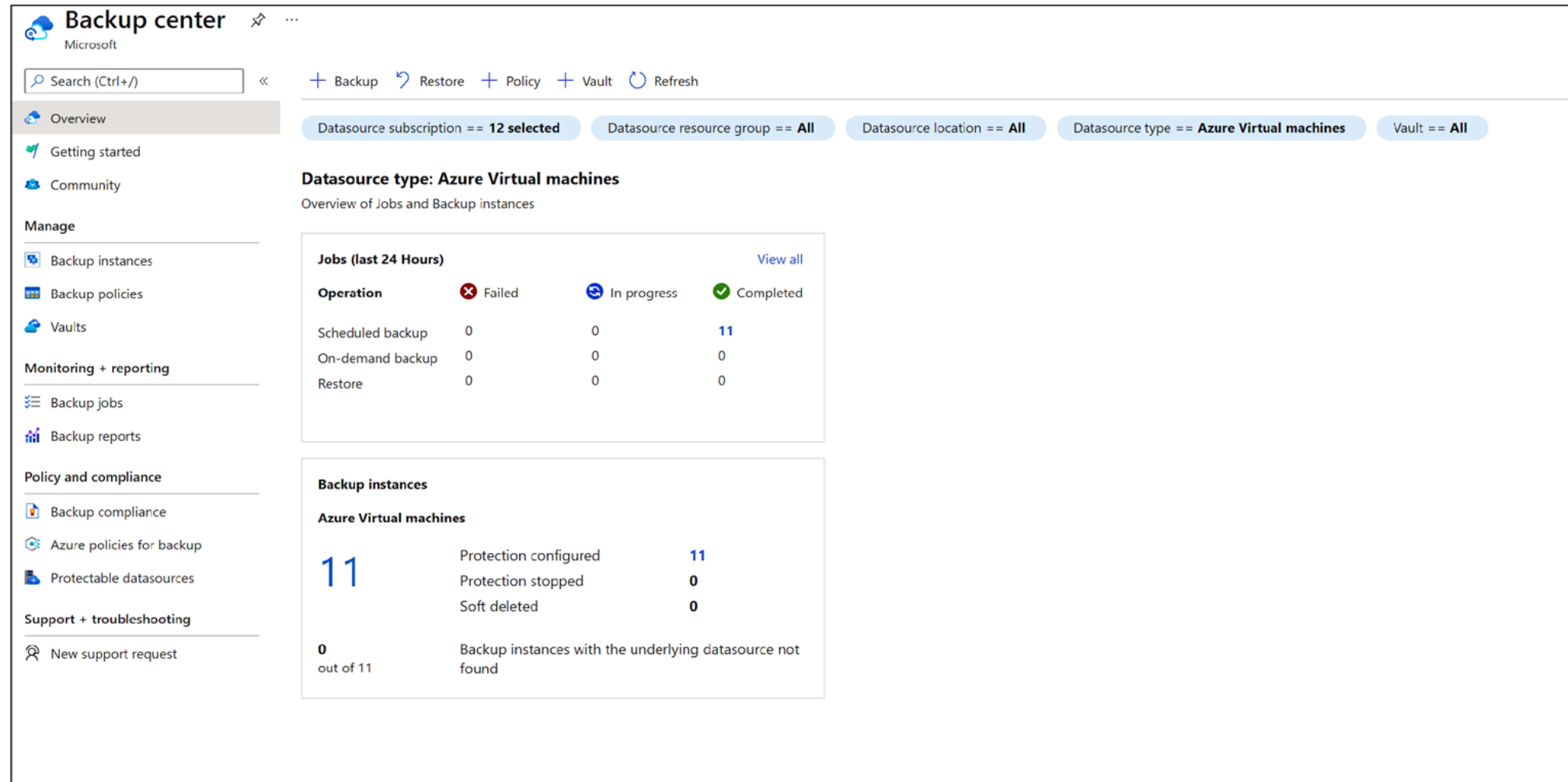
The benefits of the Backup center include:

- Single pane of glass to manage backups across a large and distributed Azure environment
- Data source-centric management focused on what is backing up
- Connected experiences with native integrations that enable management at scale

Implement Azure Backup Center

In Overview under the Backup center page, there are two tiles:

- Jobs
- Backup instances



The screenshot shows the Azure Backup Center Overview page for Azure Virtual machines. The left sidebar contains navigation links for Overview, Getting started, Community, Manage (Backup instances, Backup policies, Vaults), Monitoring + reporting (Backup jobs, Backup reports), Policy and compliance (Backup compliance, Azure policies for backup, Protectable datasources), and Support + troubleshooting (New support request). The main content area displays filters for Datasource subscription (12 selected), Datasource resource group (All), Datasource location (All), Datasource type (Azure Virtual machines), and Vault (All). Below the filters, the 'Jobs (last 24 Hours)' tile shows a table of backup operations: Scheduled backup (0 Failed, 0 In progress, 11 Completed), On-demand backup (0 Failed, 0 In progress, 0 Completed), and Restore (0 Failed, 0 In progress, 0 Completed). The 'Backup instances' tile shows 11 Azure Virtual machines with 11 Protection configured, 0 Protection stopped, and 0 Soft deleted. It also indicates 0 out of 11 Backup instances with the underlying datasource not found.

Operation	Failed	In progress	Completed
Scheduled backup	0	0	11
On-demand backup	0	0	0
Restore	0	0	0

Backup instances	Protection configured	Protection stopped	Soft deleted
11	11	0	0

0 out of 11 Backup instances with the underlying datasource not found

Source: <https://learn.microsoft.com/en-us/azure/backup/backup-center-overview>

Setup Recovery Services Vault Backup Options: Files

The recovery services vault is a storage entity in Azure that stores data. It stores backup data for various Azure services, such as IaaS VMs (Linux or Windows) and Azure SQL databases.

Azure workloads

Where is your workload running?

Azure

What do you want to backup?

Virtual machine

Virtual machine

Azure FileShare

SQL Server in Azure VM

SAP HANA in Azure VM

On-premises workloads

vmbackuptest- Backup

Recovery Services vault

Where is your workload running?

On-Premises

What do you want to backup?

Files and folders

☒ Files and folders

☐ Hyper-V Virtual Machines

☐ VMware Virtual Machines

☐ Microsoft SQL Server

☐ Microsoft SharePoint

☐ Microsoft Exchange

☐ System State

☐ Bare Metal Recovery

Step: Prepare Infrastructure

Prepare Infrastructure

Setup Recovery Services Vault Backup Options: VMs

Multiple servers can be protected using the same recovery services vault.

Azure workloads

Where is your workload running?
Azure

What do you want to backup?
Virtual machine

- Virtual machine
- Azure FileShare
- SQL Server in Azure VM
- SAP HANA in Azure VM

On-premises workloads

vmbackuptest - Backup
Recovery Services vault

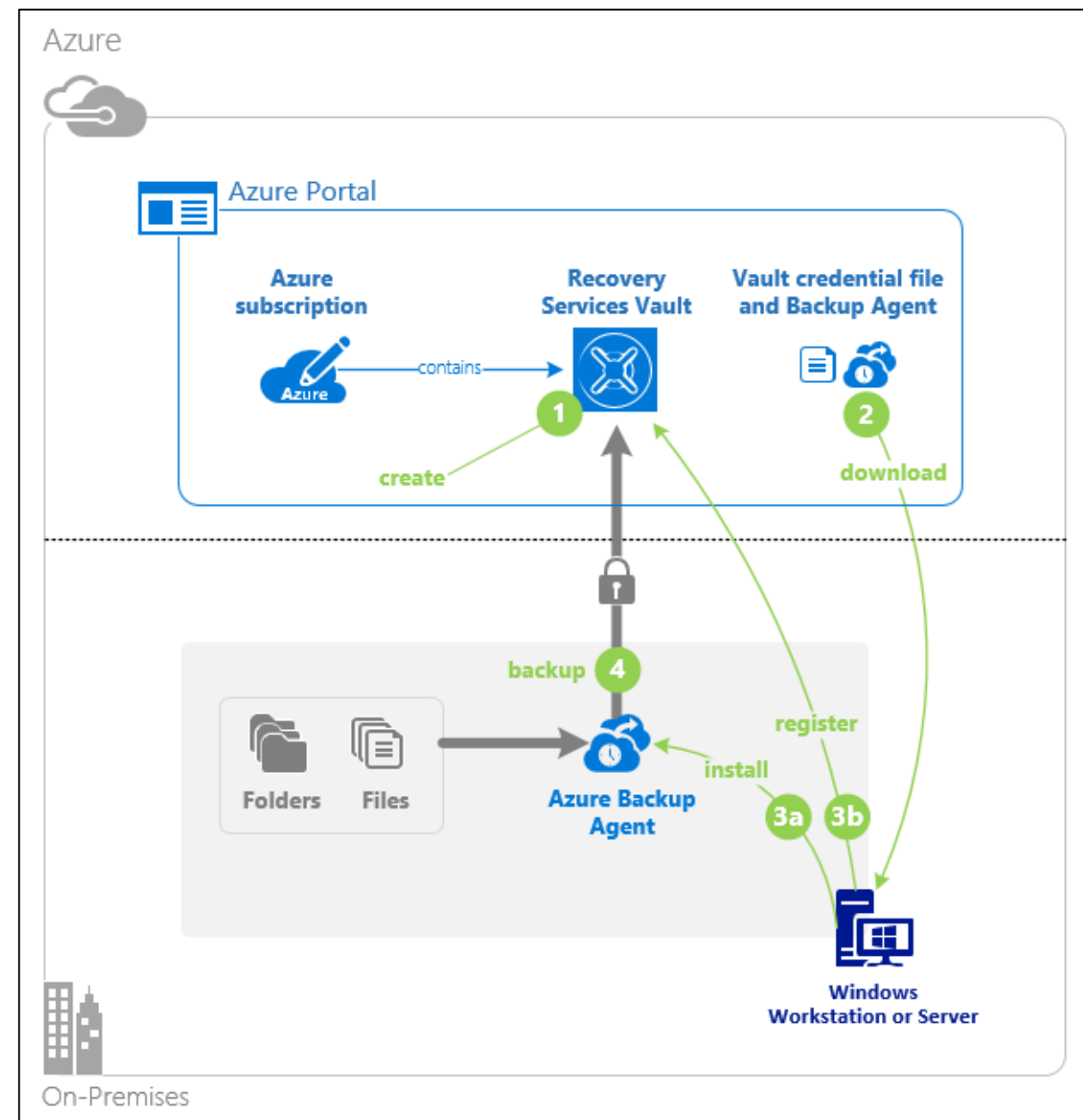
Where is your workload running?
On-Premises

What do you want to backup?
4 selected

- ☐ Files and folders
- ☒ Hyper-V Virtual Machines
- ☒ VMware Virtual Machines
- ☐ Microsoft SQL Server
- ☐ Microsoft SharePoint
- ☐ Microsoft Exchange
- ☒ System State
- ☒ Bare Metal Recovery

Step: Prepare Infrastructure
Prepare Infrastructure

Configure On-Premises File and Folder Backup



There are several steps to configure Azure backup of on-premises files and folders:

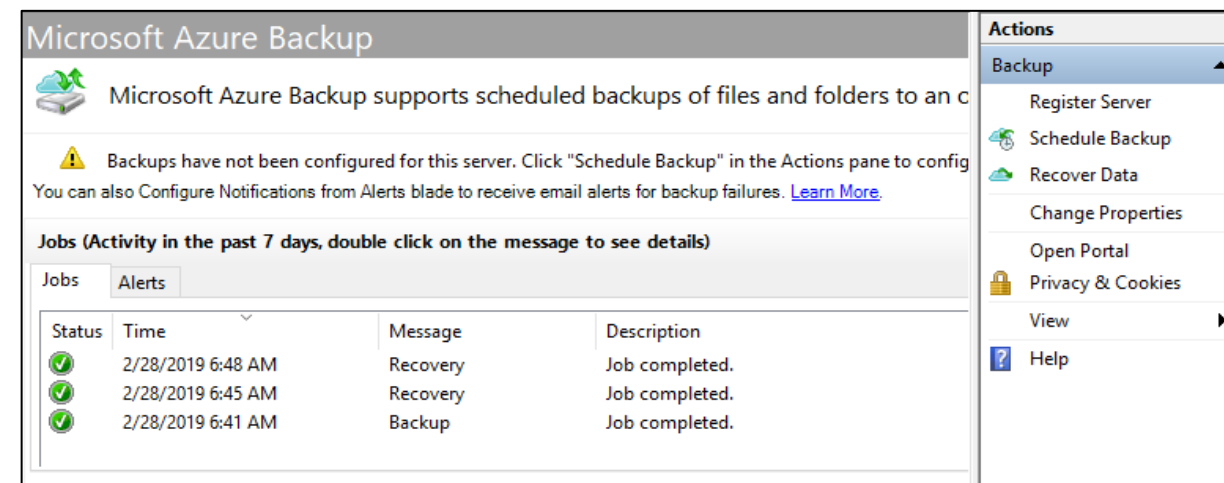
1. Create the recovery services vault
2. Download the agent and credential file
3. Install and register an agent
4. Configure the backup

Manage the Microsoft Azure Recovery Services Agent

Azure backup for files and folders relies on the Microsoft Azure Recovery Services (MARS) agent to be installed on the Windows client or server.

MARS agent is a full-featured agent that has many features, such as:

- Back up or recover files and folders on physical or virtual Windows OS (VMs can be on-premises or in Azure)
- No separate backup server is required
- Not application-aware; file, folder, and volume-level restore only
- No support for Linux



Back Up Azure File Shares



Duration: 10 Min.

Problem Statement:

You've been tasked to create an Azure file share in the Azure portal to upload, backup, and recover the files

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a recovery service vault
2. Configure backup from the recovery service vault
3. Configure backup from file share pane
4. Run an on-demand backup job



Configure Alerts for Backup

It involves setting up monitoring and notifications to ensure the health and status of the backup operations.

The screenshot displays the Microsoft Azure Backup Center dashboard. The left sidebar contains navigation links for Overview, Getting started, Community, Manage (Backup instances, Backup policies, Vaults), Monitoring + reporting (Backup jobs, Alerts (Preview), Metrics (Preview), Backup reports), Policy and compliance (Backup compliance, Azure policies for backup, Protectable datasources), and Support + troubleshooting (New support request). The main content area shows filters for Datasource subscription (Microsoft Azure Sponsorship - ModInf...), Datasource resource group (All), Datasource location (All), Datasource type (Azure Virtual machines), and Vault (All). Below the filters, there are two summary cards: 'Jobs (last 24 Hours)' and 'Backup instances'. The 'Jobs (last 24 Hours)' card shows a table of backup operations: Scheduled backup (0 Failed, 0 In progress, 0 Completed), On-demand backup (0 Failed, 0 In progress, 3 Completed), and Restore (0 Failed, 0 In progress, 0 Completed). The 'Backup instances' card shows 3 Azure Virtual machines with 3 Protection configured, 0 Protection stopped, and 0 Soft deleted. Below these cards, there is a red-bordered box containing 'Active Alerts (last 24 Hours)'. This box is divided into 'Azure Virtual machines' and 'Global alerts'. Both sections show 0 Security alerts and 0 Configured alerts, with a breakdown by severity: Sev0 (0/0), Sev1 (0/0), and Sev2 (0/0). A link at the bottom of the red box says 'Click here to learn more about the alerts displayed in Backup center'.

Operation	Failed	In progress	Completed
Scheduled backup	0	0	0
On-demand backup	0	0	3
Restore	0	0	0

Alert Type	Sev0	Sev1	Sev2
Azure Virtual machines	0/0	0/0	0/0
Global alerts	0/0	0/0	0/0



Configure Alerts for Backup

There are two primary categories of built-in notifications offered by Azure Backup:

Security alerts



Configured alerts



Security alerts

High-severity (Sev 0) alerts are triggered by Azure for operations like deleting backups or deactivating soft deletion in a vault.

Configured alerts

Azure generates high-severity (Sev 0) alarms when users do operations like deleting backups or turning off soft deletion in a vault.

Configure Alerts for Backup

Types of notifications used in the context of data backup and recovery are:

Backup
alerts

They notify about the success or failure of scheduled backup jobs.

Restore
alerts

They provide information about the success or failure of restoration operations.

Replicatio
n alerts

They inform about the replication health, ensuring recovery points meet the set SLA.



Benefits of Azure Monitor Alerts

Broad notification capabilities

They provide quick backup alert setups across channels like email, ITSM, and webhooks without custom integrations.

Selective notifications

They can tailor notifications to specific scenarios and set alerts for test subscriptions.

Enhanced alert management

They offer a specialized alert management experience through the backup center.

Programmatic management alert

They offer REST APIs that allow for non-portal alert and notification management.

Unified alert management

They ensure consistent alert management across Azure services, allowing backup alerts to be handled.

Protect Virtual Machine Data

The user can protect the data by taking backups at regular intervals. There are several backup options available for VMs, depending on the use case.

Snapshots

Manages snapshots that provide a quick and simple option for backing up VMs that use managed disks

Azure backup

Supports application-consistent backups for both Windows and Linux VMs

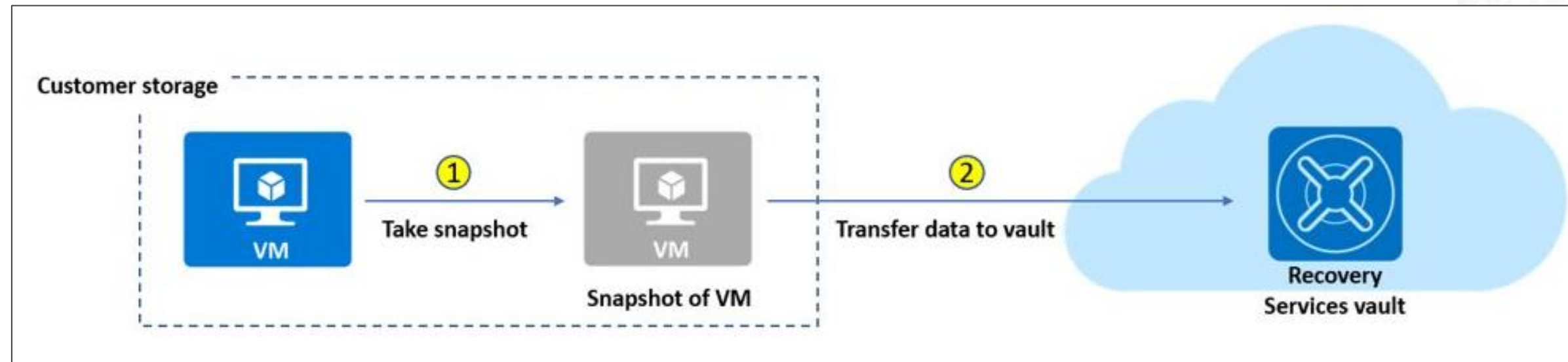
Azure site recovery

Protects VMs from a major disaster scenario when a whole region experiences an outage

Create Virtual Machine Snapshots

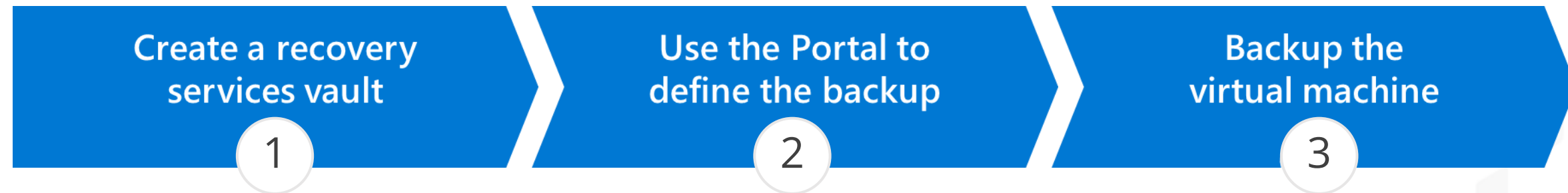
Steps for Azure backup job:

- Uses snapshots taken as part of a backup job
- Reduces recovery wait times: Do not wait for data transfer to the vault to finish
- Configures instant restore retention (1 to 5 days)



Backup Virtual Machines

The Backup of Azure Virtual Machines using Azure Backup is easy and follows a simple process:



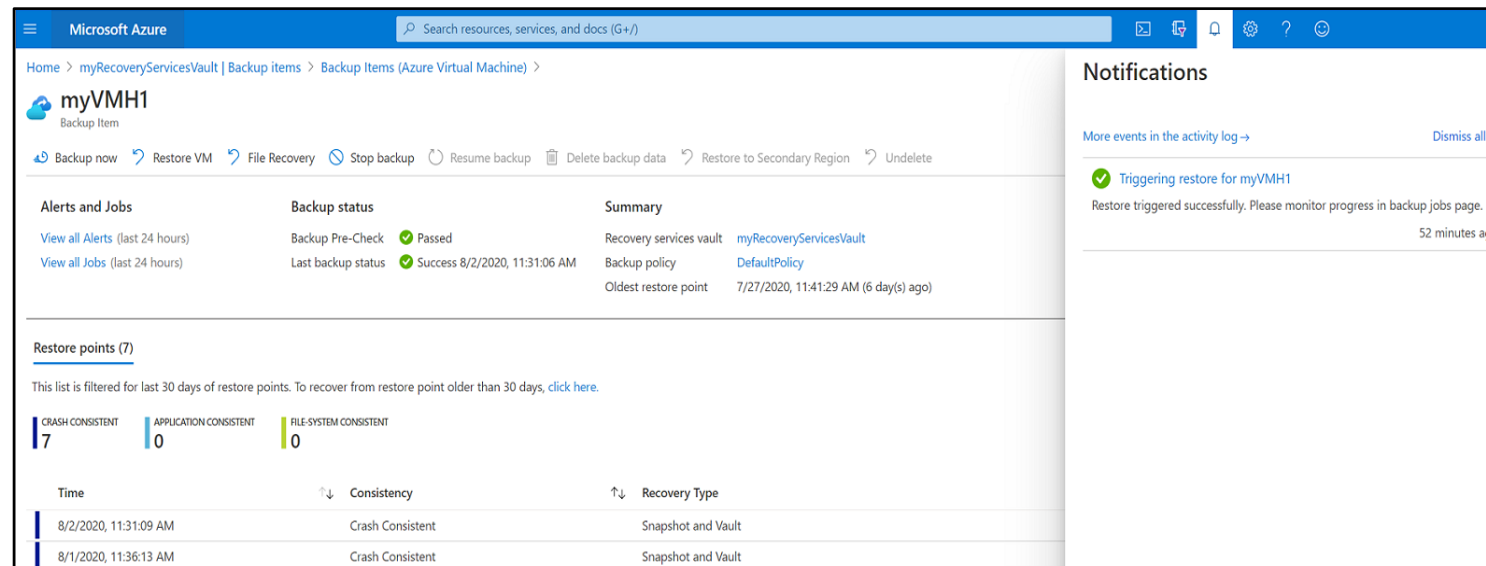
1. Use the recovery services vault in the region where the user is performing their Virtual Machine backups and choose a replication strategy for the vault.

2. Take snapshots (recovery points) of the data at defined intervals. These snapshots are stored in recovery services vaults.

3. Install the Azure VM agent on the Azure Virtual Machine for the backup extension to work.

Restore Virtual Machines

Once the virtual machine snapshots are put safely in the recovery services vault, it is simple to recover them.

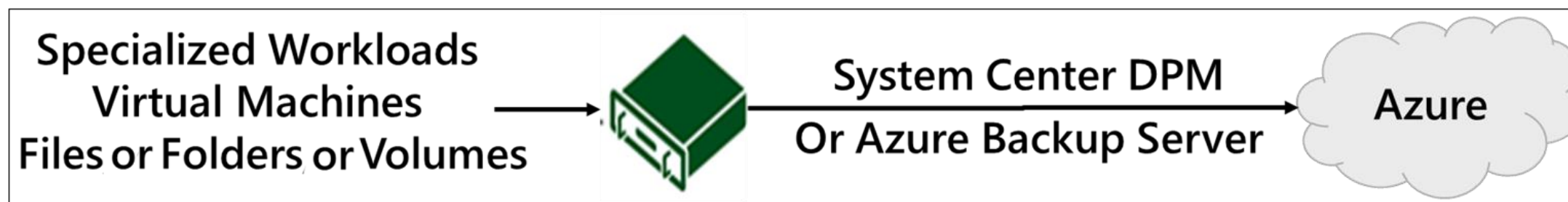


- The backup service creates a job for tracking the restore operation, once the user triggers the restore operation.
- The backup service also creates and temporarily displays notifications so that the user can monitor how the backup is proceeding.

Implement Azure Backup Server

There are several advantages of backing up machines and apps to MABS or DPM storage and then backing up DPM or MABS storage to a vault, namely:

- App-aware backups, file or folder or volume backups, and machine-state backups (bare-metal, system state)
- Each machine runs the DPM or MABS protection agent, and the MARS agent runs on the MABS or DPM
- Flexibility and granular scheduling options
- Manage backups for multiple machines in a protection group



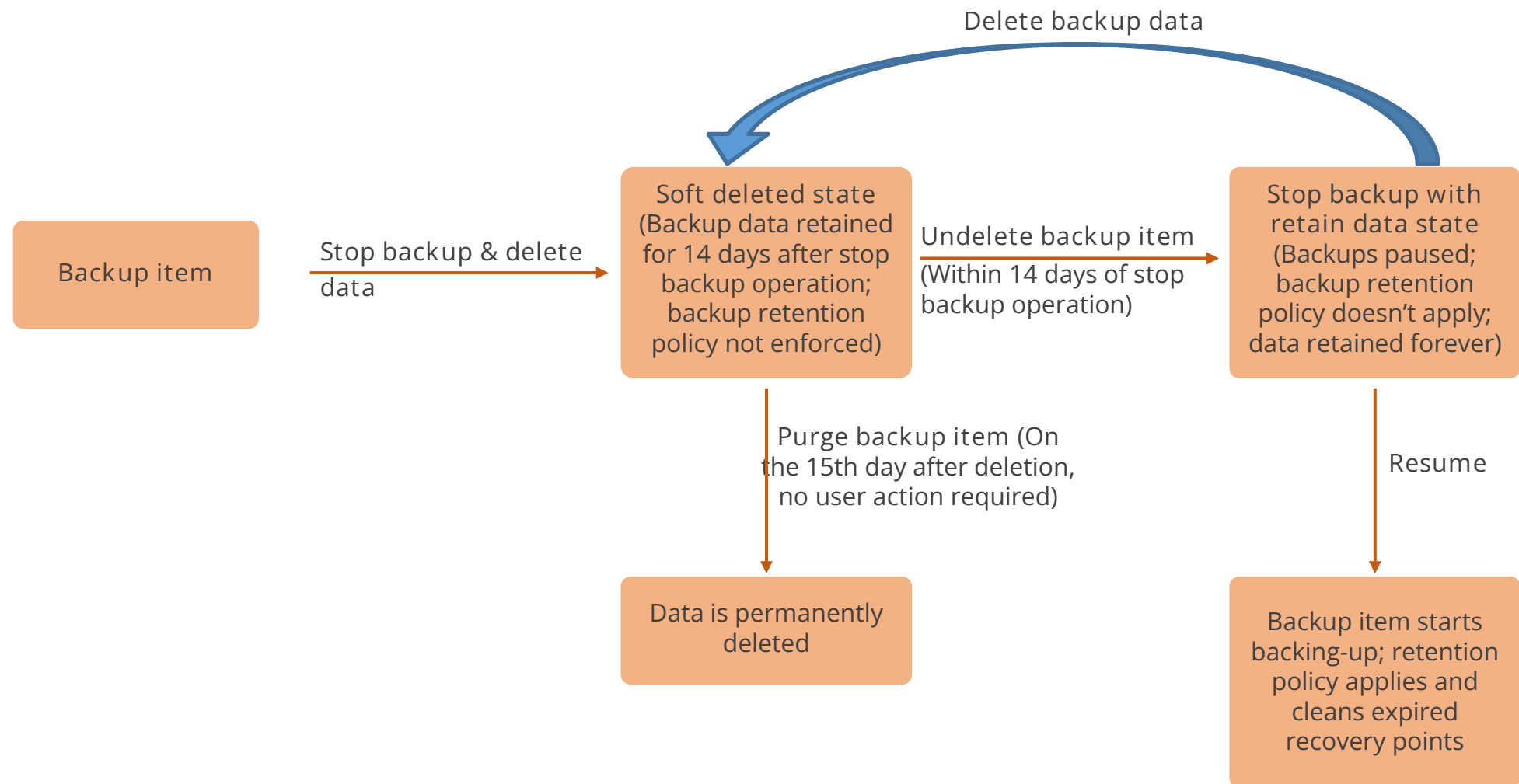
Compare Backup Options

The Backup options are compared below:

Component	Benefits	Limits	Protects	Backup Storage
Azure Backup (MARS) agent	• Backup files and folders on physical or virtual Windows OS • No separate backup server required	• Backup 3x per day • Not application aware • File-, folder-, and volume-level restore only • No support for Linux	• Files • Folders	• Recovery services vault
Azure Backup Server (MABS)	• App-aware snapshots • Full flex during backups • Recovery granularity • Linux support on Hyper-V and VMware VMs • Backup and restore VMware VMs • Does not require a System Center license	• Cannot backup Oracle workloads • Always requires live Azure subscription • No support for tape backup	• Files • Folders • Volumes • VMs • Applications • Workloads	• Recovery Services vault • Locally attached disk

Manage Soft Deletes

It is a feature that ensures that backup data remains protected against inadvertent deletions.



- Backup data is retained for an additional 14 days.
- Soft deleted backup items can be restored through an undelete operation.
- This capability is available for storage account containers and file shares.
- It is inherently integrated into all recovery services vaults

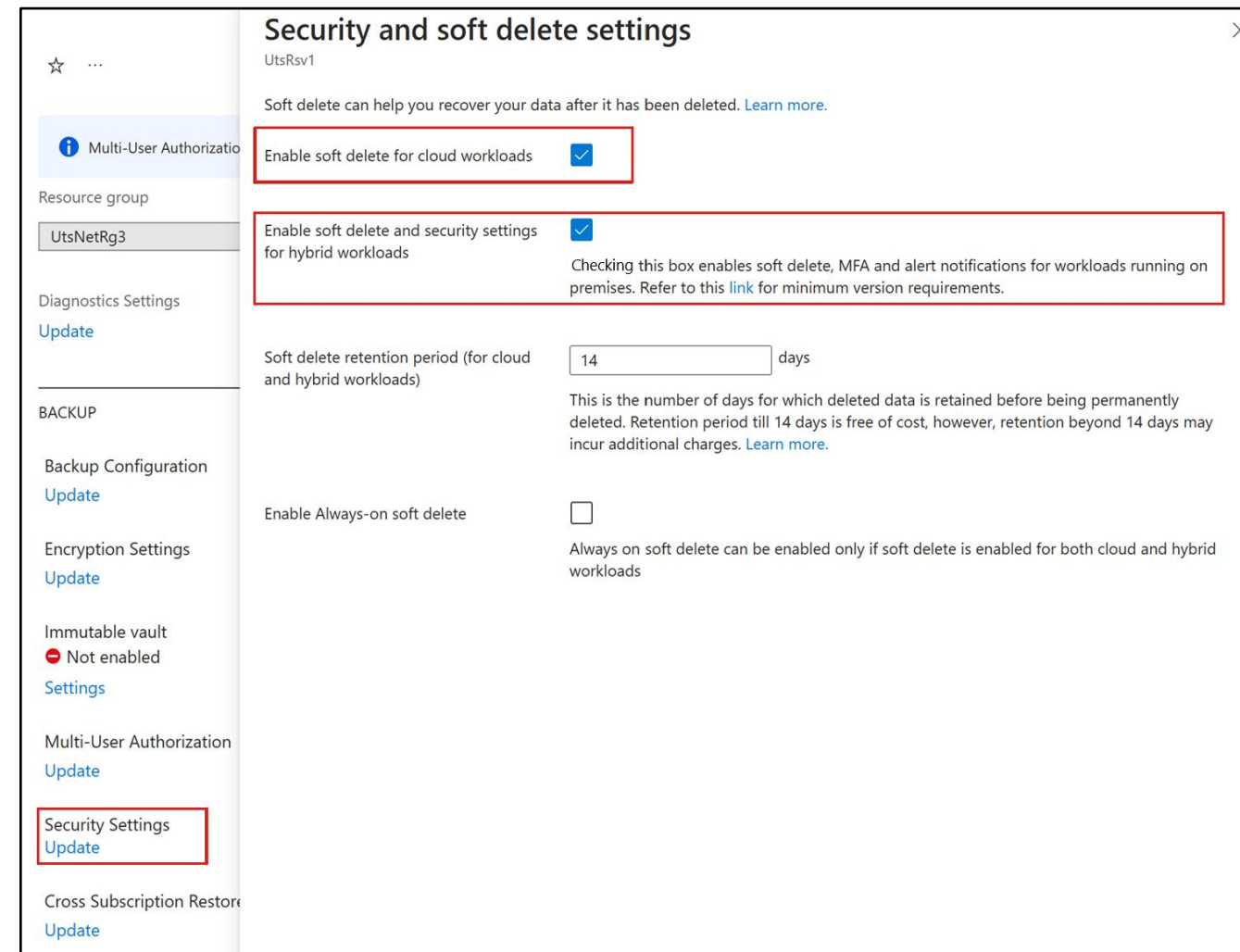
Enabling and Disabling Soft Delete



- Soft delete is activated once; the always-on soft delete feature is permanent and cannot be turned off.
- Backup data previously marked as soft deleted remains as such even after disabling the feature; to permanently remove it, you need to restore (undelete) and then redelete it.

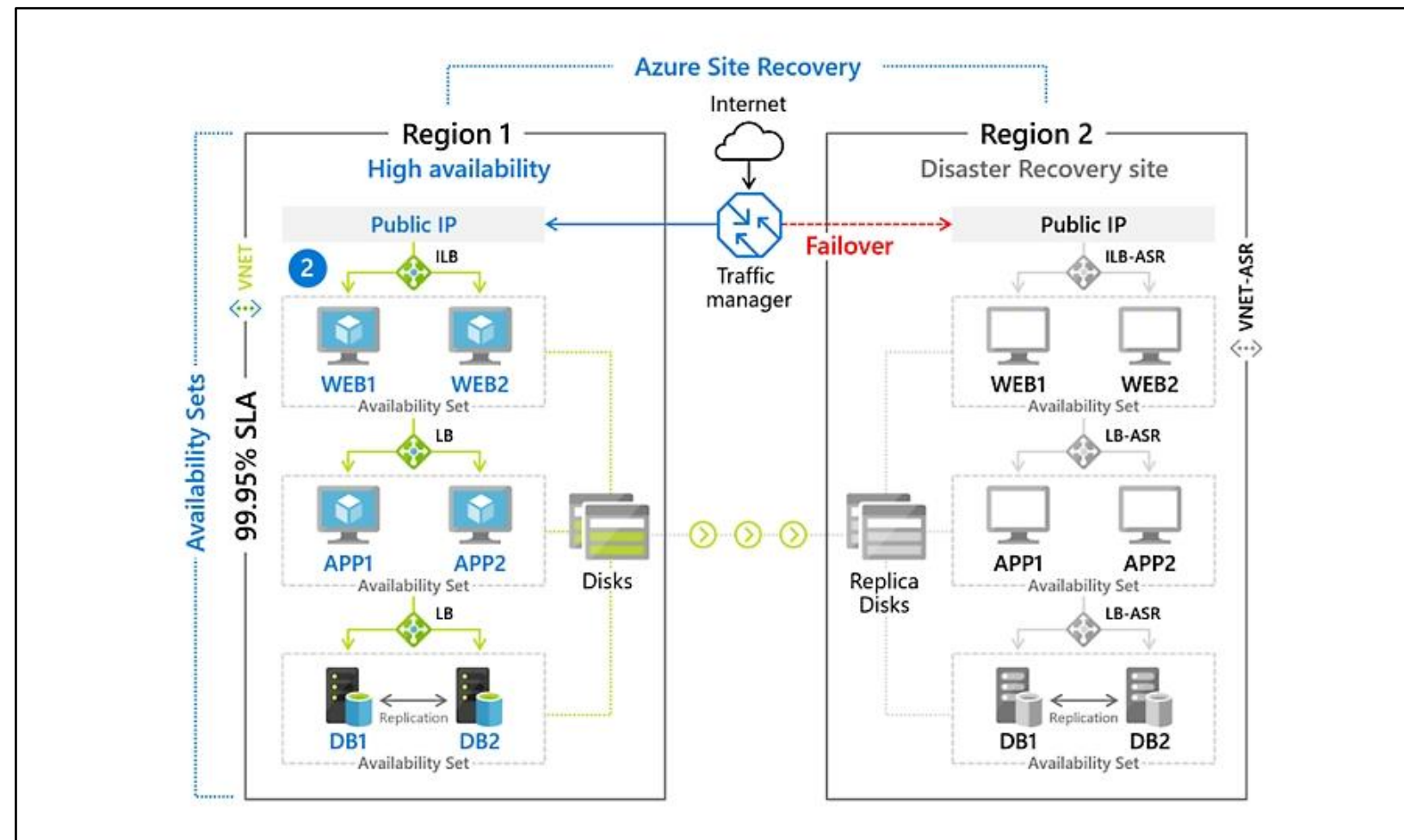
Disabling Soft Delete Using Azure Portal

- Navigate to the vault, then select Settings > Properties
- Choose Security Settings Update from the Properties section
- Within the Security and soft delete settings, deselect the check boxes associated with soft delete



Azure Site Recovery

It helps organizations protect their on-premises or cloud-based workloads by providing continuous replication and automated recovery options.



It ensures business continuity by enabling replication of virtual machines, physical servers, and even entire data centers to Azure, facilitating fast and efficient recovery during unforeseen outages or disasters.

Azure Site Recovery: Features

Automated recovery

Azure site recovery enables automated recovery of replicated workloads in a disaster or outage.

Application consistency

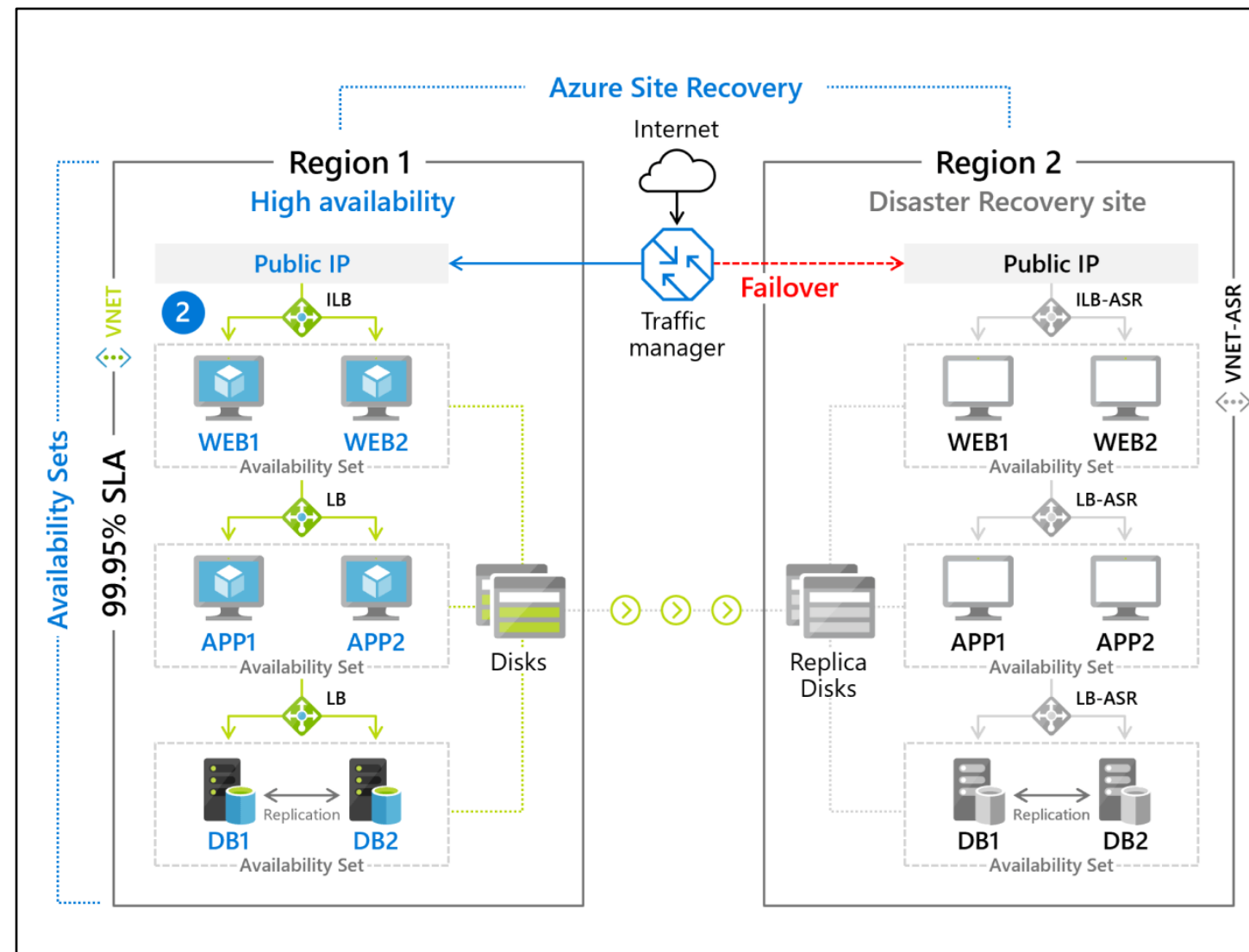
It ensures application consistency during failover by coordinating replication and recovery of multitier applications.

Cross-platform support

It ensures enabling replication and recovery for a variety of operating systems and applications.

Implement Azure Site Recovery

Site recovery keeps business apps and workloads running during outages, thus helping in ensuring business continuity. Replicate scenarios include:



- Replicate Azure VMs from one Azure region to another
- Replicate on-premises VMware VMs, Hyper-V VMs, physical servers (Windows and Linux), Azure stack VMs to Azure
- Replicate AWS Windows instances to Azure
- Replicate on-premises VMware VMs, Hyper-V VMs managed by system center VM, and physical servers to a secondary site

Back Up Virtual Machine from VM Settings



Duration: 10 Min.

Problem Statement:

You've been tasked to back up Azure Virtual Machines using Azure backup service by creating the backup policy.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Sign in to the Azure portal
2. Select the VM to backup
3. On the VM menu, select Backup
4. Add a recovery service vault
5. Choose a backup policy
6. Enable backup



Administer Monitoring

Describe Azure Monitor Key Capabilities

- Provides core monitoring for Azure services
- Collects metrics, activity logs, and diagnostic logs
- Can be used for time-critical alerts and notifications



Monitor & Visualize Metrics

Metrics are numerical values available from Azure Resources helping you understand the health, operation & performance of your systems.

[Explore Metrics](#)



Query & Analyze Logs

Logs are activity logs, diagnostic logs and telemetry from monitoring solutions; Analytics queries help with troubleshooting & visualizations.

[Search Logs](#)



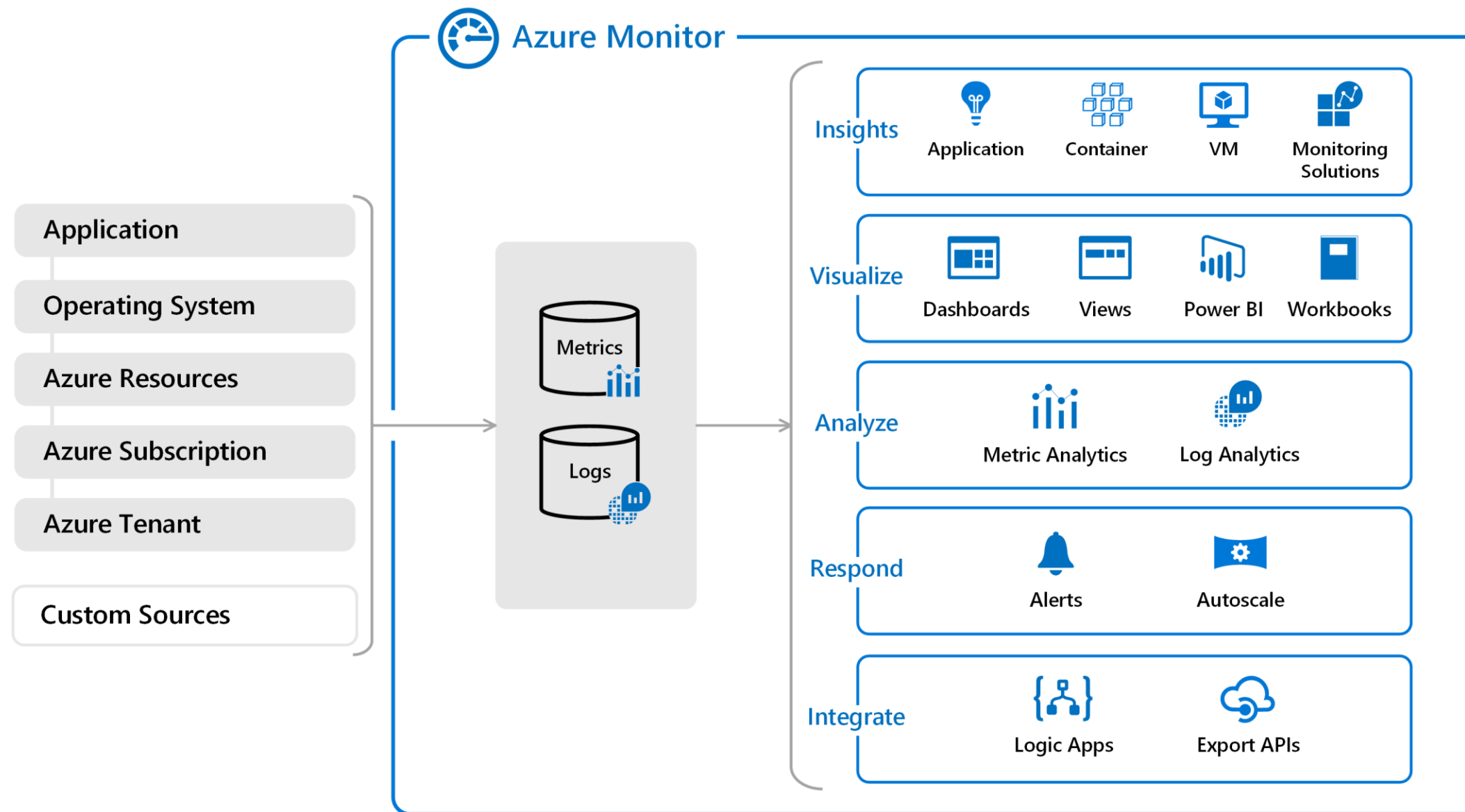
Setup Alert & Actions

Alerts notify you of critical conditions and potentially take corrective automated actions based on triggers from metrics or logs.

[Create Alert](#)

Understand Azure Monitor Components

Azure provides services that deliver a comprehensive solution for collecting, analyzing, and performing data. Some of the functions that Azure monitor performs are presented in the diagram below.

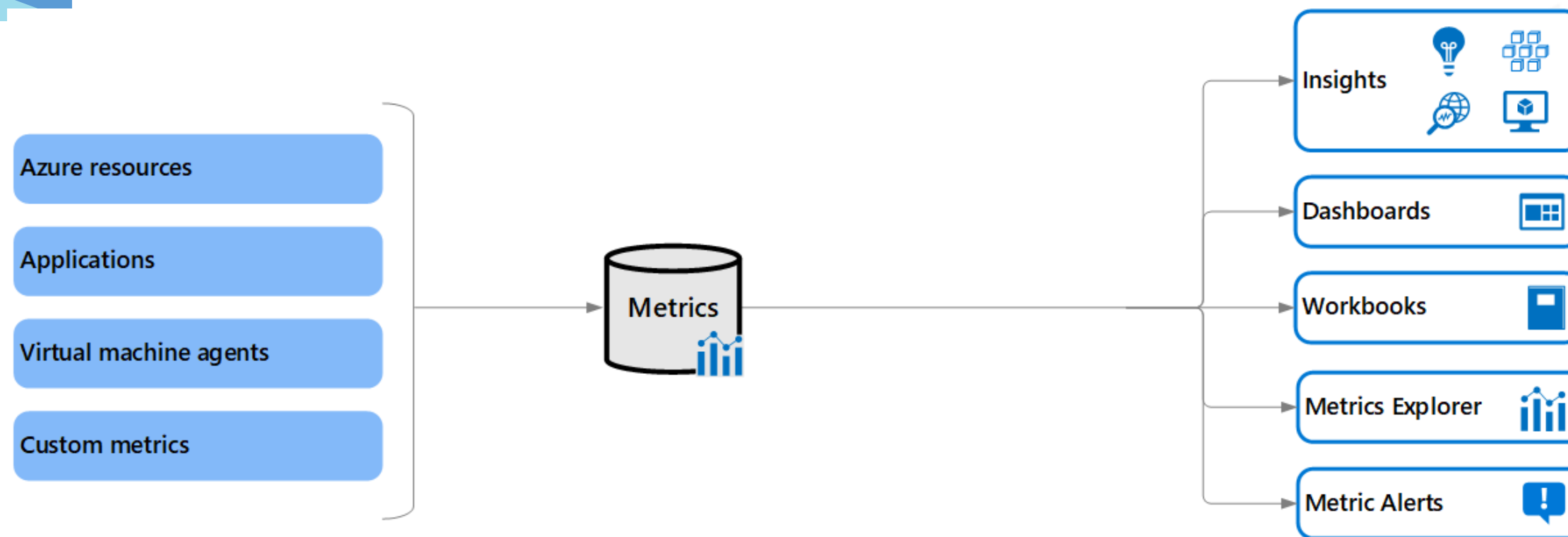


Source: <https://learn.microsoft.com/en-us/dotnet/architecture/cloud-native/azure-monitor>

Define Metrics and Logs

All data gathered by Azure monitor fits into two fundamental types: metrics and logs.

- Metrics are numerical values that describe some aspect of a system at a point in time.
- They are lightweight and capable of supporting almost real-time scenarios.

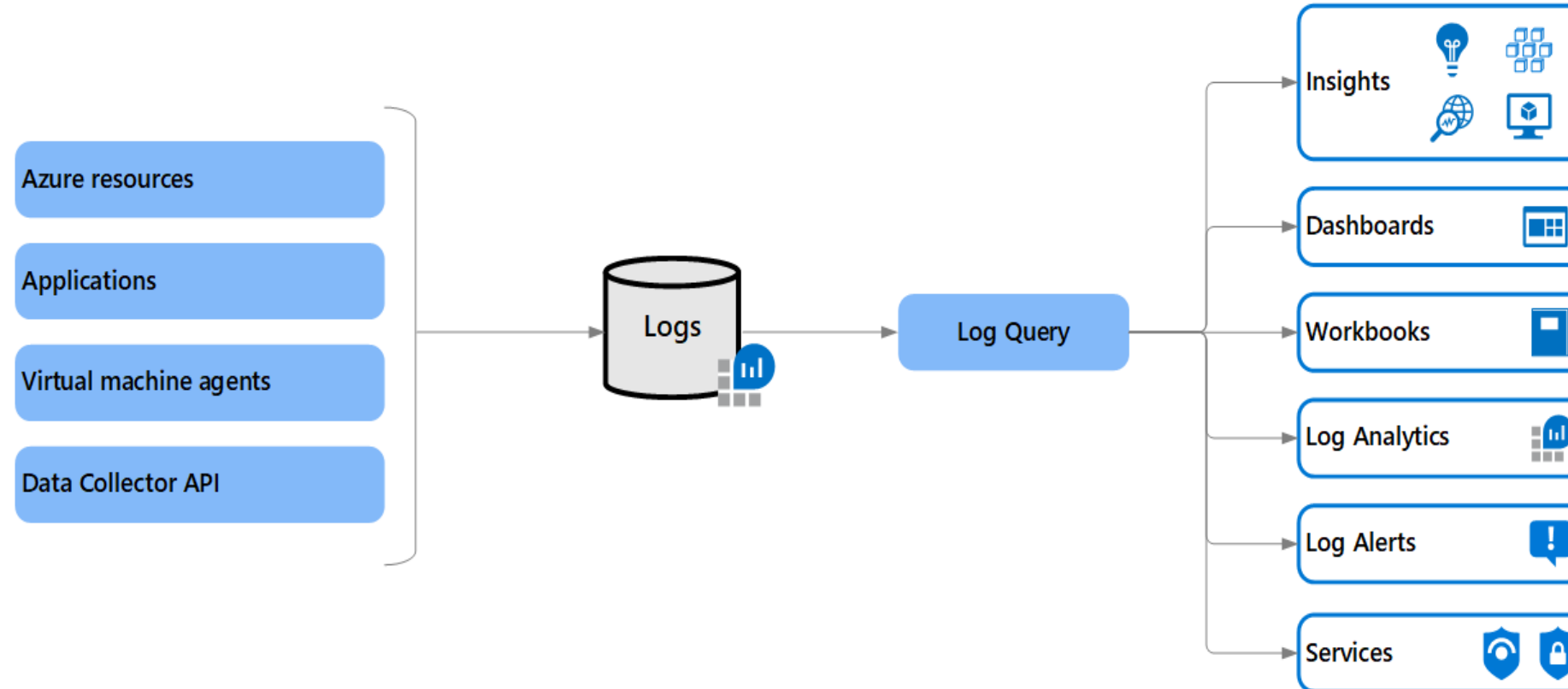


Source: <https://learn.microsoft.com/en-us/windows-server/storage/storage-spaces/configure-azure-monitor>

Define Metrics and Logs

All data gathered by Azure Monitor fits into either of the two fundamental types: metrics and logs.

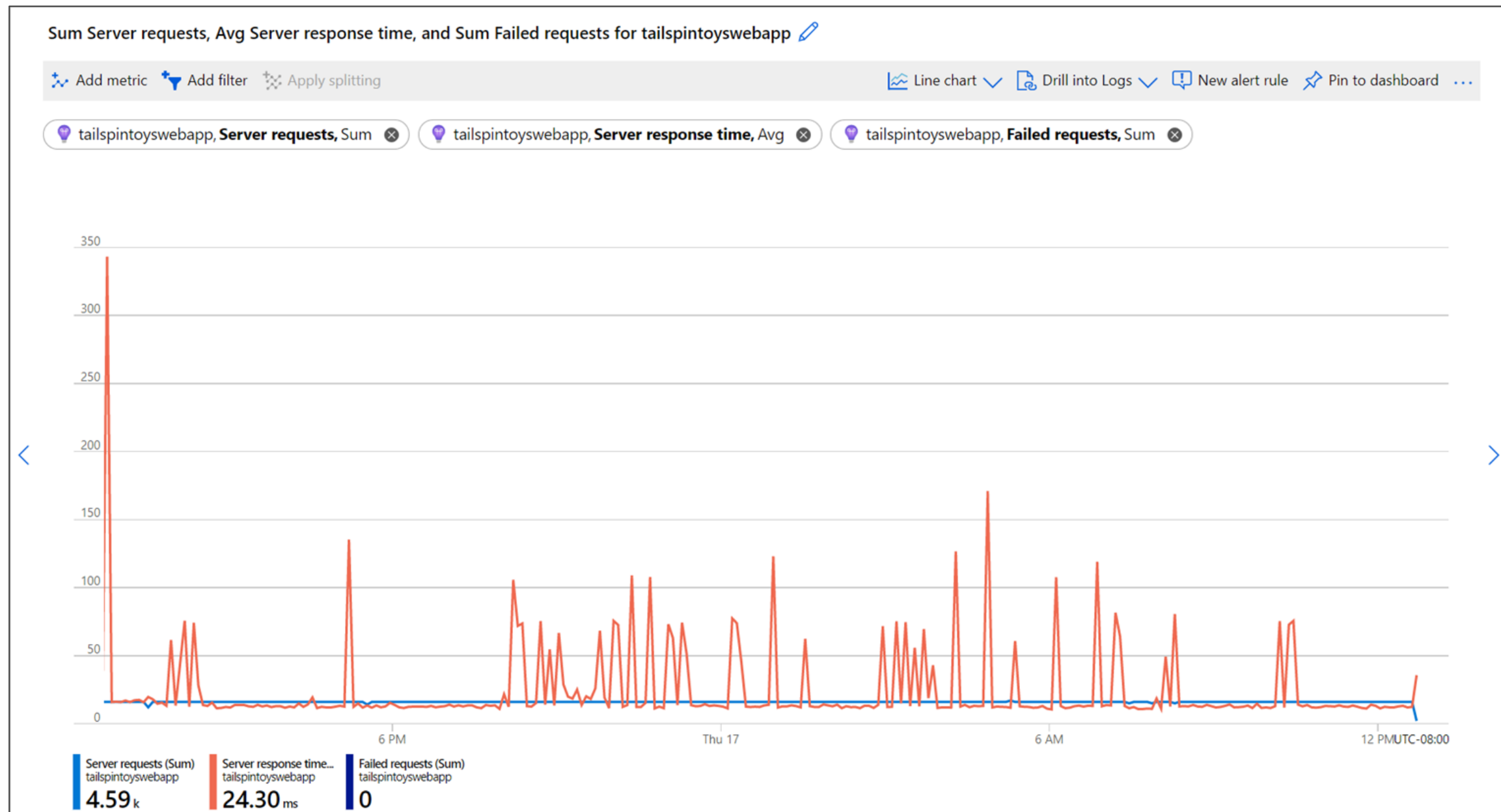
- Logs contain different kinds of data organized into records with different sets of properties for each type.
- Telemetry (events, traces) and performance data can be combined for analysis.



Source: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/data-platform-logs>

Define Metrics and Logs

Within Azure Monitor, a tool called metrics explorer enables interactive analysis of data stored in metric databases.



Source: <https://learn.microsoft.com/en-us/azure/azure-monitor/essentials/data-platform-metrics>

Define Metrics and Logs

The following is a screenshot of a log:

The screenshot shows the Microsoft Monitor - Logs interface. The left sidebar contains navigation options: Overview, Activity log, Alerts, Metrics, Logs (selected), Service Health, Insights, Applications, Virtual Machines (preview), Containers, Network, More, Settings, Diagnostics settings, Autoscale, Support + Troubleshooting, Usage and estimated costs, Advisor recommendations, and New support request. The main area displays a query titled 'New Query 1*' with a 'Run' button and a 'Time range: Last 24 hours' filter. The query is as follows:

```
// Availability rate
// Calculate the availability rate of each connected computer
Heartbeat
// bin_at is used to set the time grain to 1 hour, starting exactly 24 hours ago
| summarize heartbeatPerHour = count() by bin_at(TimeGenerated, 1h, ago(24h)), Computer
| extend availablePerHour = iff(heartbeatPerHour > 0, true, false)
| summarize totalAvailableHours = countif(availablePerHour == true) by Computer
| extend availabilityRate = totalAvailableHours*100.0/24
```

The results are displayed in a table with columns: Computer, totalAvailableHours, and availabilityRate. The table shows 62 records, with the first 10 rows displayed. The status bar indicates 'Completed. Showing results from the last 24 hours.' and '00:00:00.424 62 records'.

Computer	totalAvailableHours	availabilityRate
App04	24	100
retailEUS8	24	100
aks-agentpool-40719753-2	24	100
App06	24	100
deletethisvm	24	100
Data06	24	100
ContosoWeb	24	100
ContosoSQLSrv2.ContosoRetail.com	24	100
OnPremise-12S	24	100
Office365	24	100

Source: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/data-platform-logs>

Identify Data Types

Azure Monitor can collect data from various sources. The user can monitor data for their applications in tiers starting from the application, any software, and services it relies on, right down to the platform.

Azure Monitor collects data from each of the subsequent tiers:

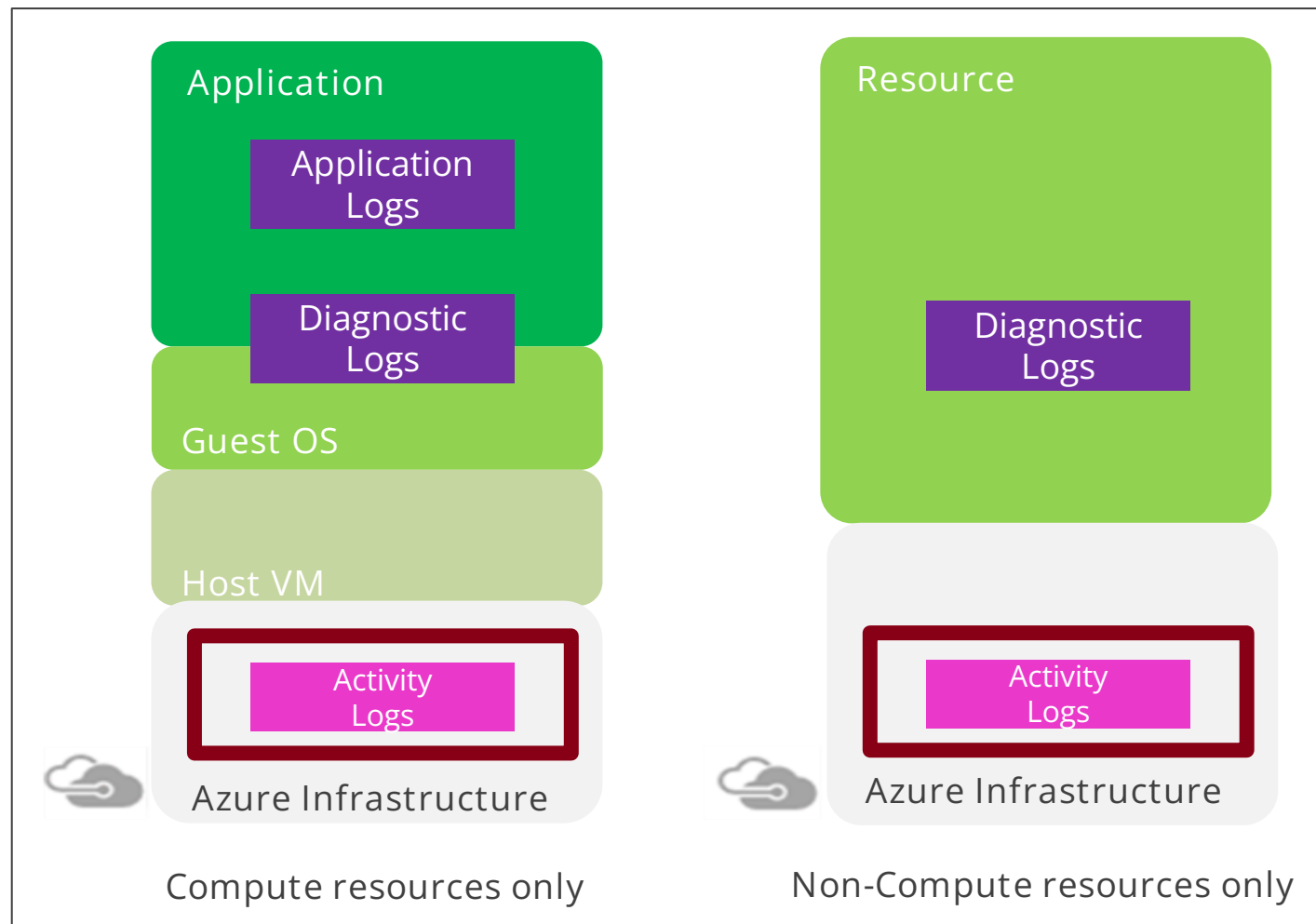
- Application monitoring data: Performance and functionality of the code written, regardless of its platform
- Guest OS monitoring: Azure, another cloud, or on-premises
- Azure resource monitoring
- Azure subscription monitoring: Operation and management of an Azure subscription as well as data about the health and operation of Azure

Azure tenant monitoring: Operation of tenant-level Azure services, such as Azure active directory



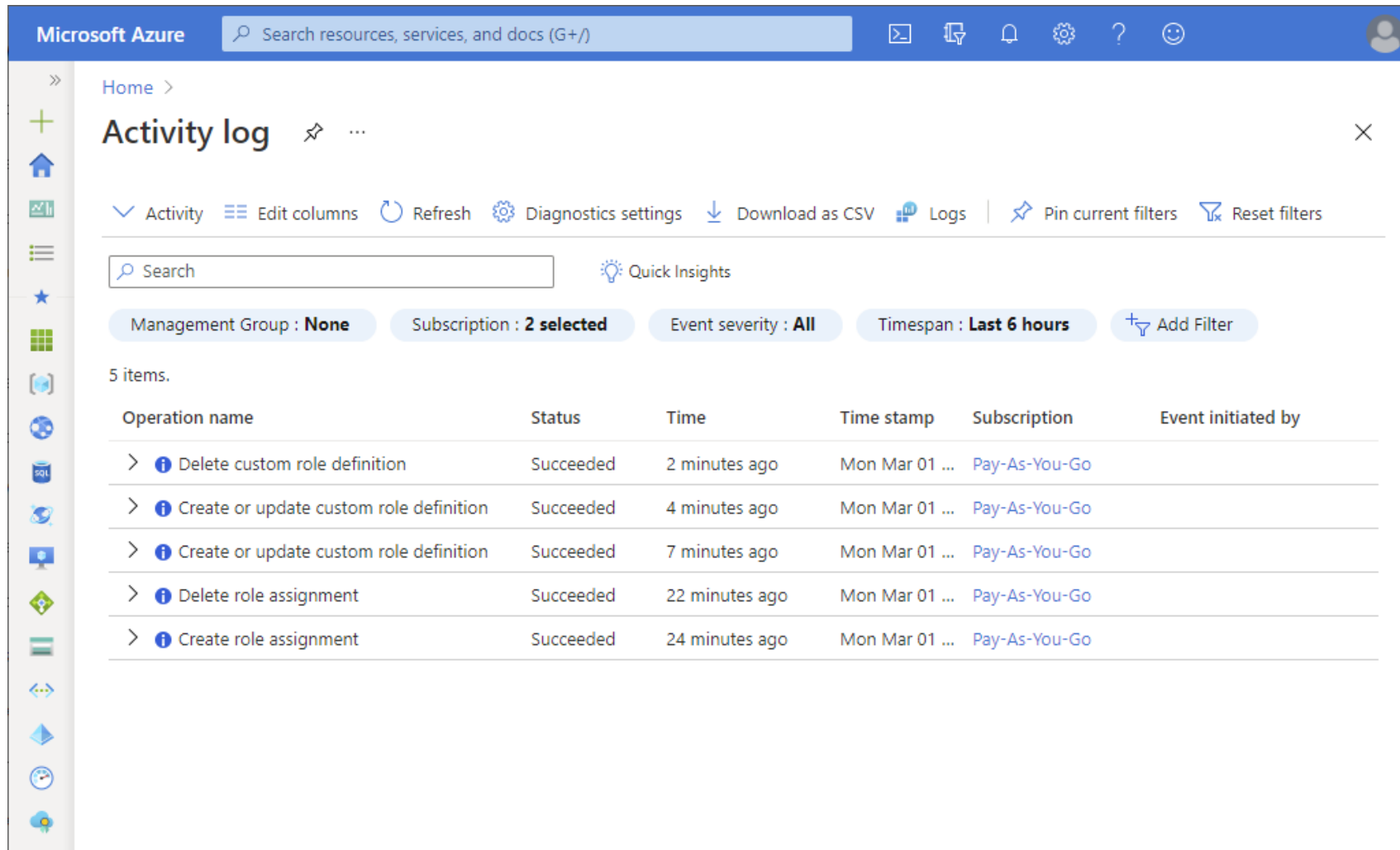
Describe Activity Log Events

It is a subscription log in which a user determines the what, who, and when for any write operations (PUT, POST, and DELETE) taken on the resources in their subscription. Its features include:



- Send data to log analytics for advanced search and alerts
- Query or manage events in the portal, PowerShell, CLI, and REST API
- Stream information to Event Hub
- Archive data to a storage account
- Analyze data with Power BI

Query the Activity Log



The screenshot shows the Microsoft Azure portal's Activity Log. The top navigation bar includes the Microsoft Azure logo, a search bar, and various icons. The left sidebar contains navigation links. The main content area is titled 'Activity log' and includes a search bar, a 'Quick Insights' button, and filter buttons for 'Management Group: None', 'Subscription: 2 selected', 'Event severity: All', and 'Timespan: Last 6 hours'. Below these filters, it says '5 items.' and displays a table of activity log entries.

Operation name	Status	Time	Time stamp	Subscription	Event initiated by
> Delete custom role definition	Succeeded	2 minutes ago	Mon Mar 01 ...	Pay-As-You-Go	
> Create or update custom role definition	Succeeded	4 minutes ago	Mon Mar 01 ...	Pay-As-You-Go	
> Create or update custom role definition	Succeeded	7 minutes ago	Mon Mar 01 ...	Pay-As-You-Go	
> Delete role assignment	Succeeded	22 minutes ago	Mon Mar 01 ...	Pay-As-You-Go	
> Create role assignment	Succeeded	24 minutes ago	Mon Mar 01 ...	Pay-As-You-Go	

In the Azure portal, the user filters the Activity Log.

- Filter by the Management Group, Subscription, Timespan, and Event severity
- Add a filter, like an event category (security, recommendations, and alerts)
- Pin current filters and download them as CSV

Create a Disaster Recovery for Azure VMs



Duration: 10 Min.

Problem Statement:

You've been tasked to create an Azure site recovery for virtual machines by enabling VM replication to another region

ASSISTED PRACTICE

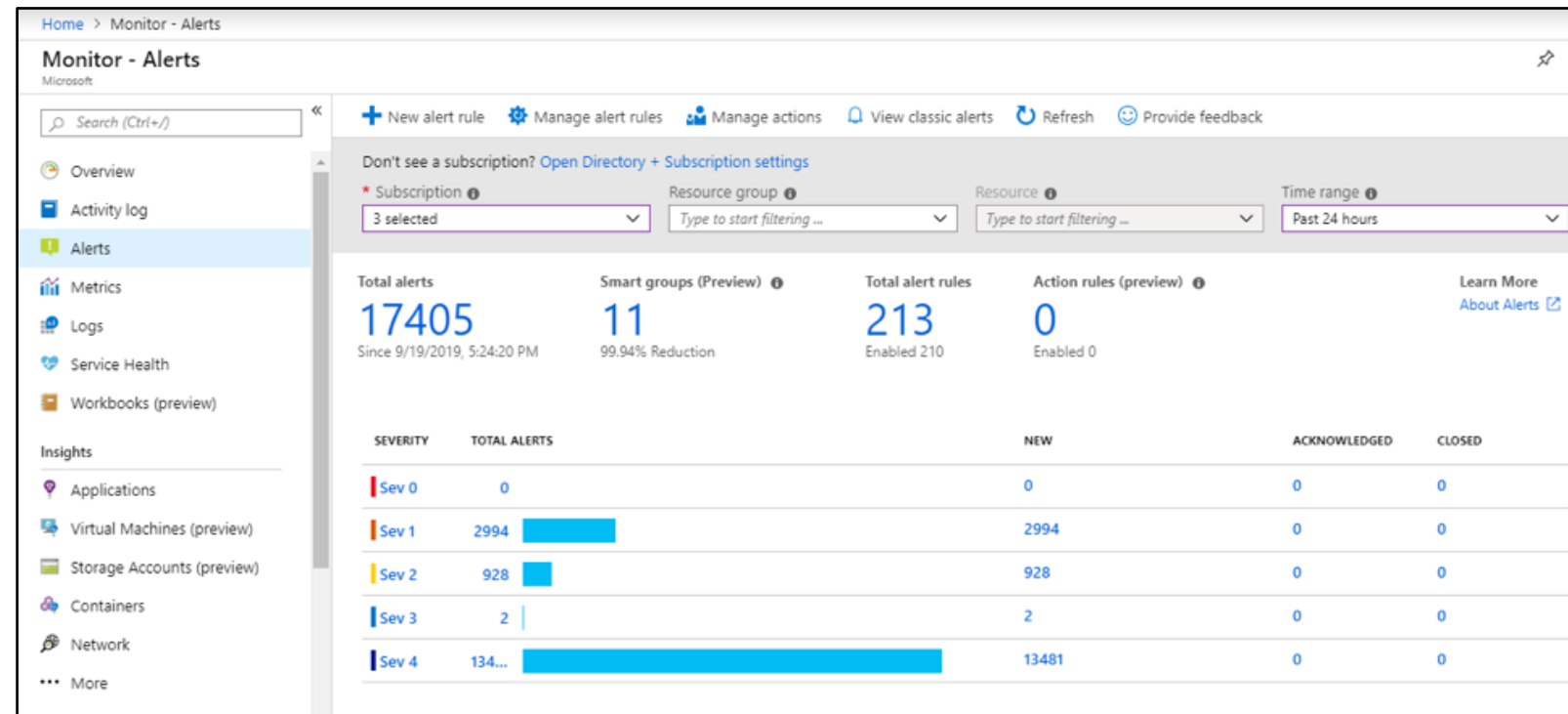
Assisted Practice: Guidelines

Steps to be followed:

1. Locate the recovery services vault blade
2. Select Enable site recovery
3. Select the virtual machines to be replicated in the destination location
4. Enable replication



Manage Azure Monitor Alerts



Monitor alert experience has many benefits.

- Unified authoring experience
- Displayed by severity
- Categorized by new, acknowledged, and closed

Source: <https://www.cloudiqtech.com/building-a-unified-monitoring-dashboard-with-azure-alerts-a-quickstart-guide/>

Create Alert Rules

Alerts proactively notify when important conditions are found in the monitoring data. Alerts contain alert rules, action groups, and monitor conditions.

Home > Monitor | Alerts >

Create an alert rule

Scope Condition Actions Details Tags Review + create

Create an alert rule to identify and address issues when important condition

+ Select scope

Resource

No resource selected yet

Select a resource

Browse Recent

Subscription: 4 subscriptions | Resource types: All resource types | Locations: All locations

Search to filter items...

Resource	Resource type	Location
☐ > Contoso Hotels	Subscription	-
☐ > Contoso Hotels Tenant - Production	Subscription	-
☐ > Contoso IT - Retail - Prod	Subscription	-
☐ > prodTeamTesting	Subscription	-

- Scope: Target selection, alert criteria, and alert logic
- Alert rule details: Name, description, and severity (0 to 4)
- Action group: Notify the team via email and text messages or automate actions using webhooks and runbooks

Create Action Groups

An action group could be a group of notification preferences defined by the owner of an Azure subscription.

Notifications

Choose how to get notified when the action group is triggered. This step is optional.

Notification type ⓘ	Name ⓘ	Selected ⓘ
Email Azure Resource Manager Role Email/SMS message/Push/Voice		

Configure the method in which users will be notified when the action group triggers

Actions

Choose which actions are performed when the action group is triggered. This step is optional.

Action type ⓘ	Name ⓘ	Selected ⓘ
Automation Runbook Azure Function Event Hub ITSM Logic App Secure Webhook Webhook		

Configure the method in which actions are performed when the action group triggers

Source: <https://media.geeksforgeeks.org/wp-content/uploads/20221106003046/monitor55.png>

Use Activity Log Alerts to Alert Azure Infrastructure



Duration: 10 Min.

Problem Statement:

Create activity log alerts to notify when some event happens on an Azure resource or some planned maintenance happens on the Azure platform.

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

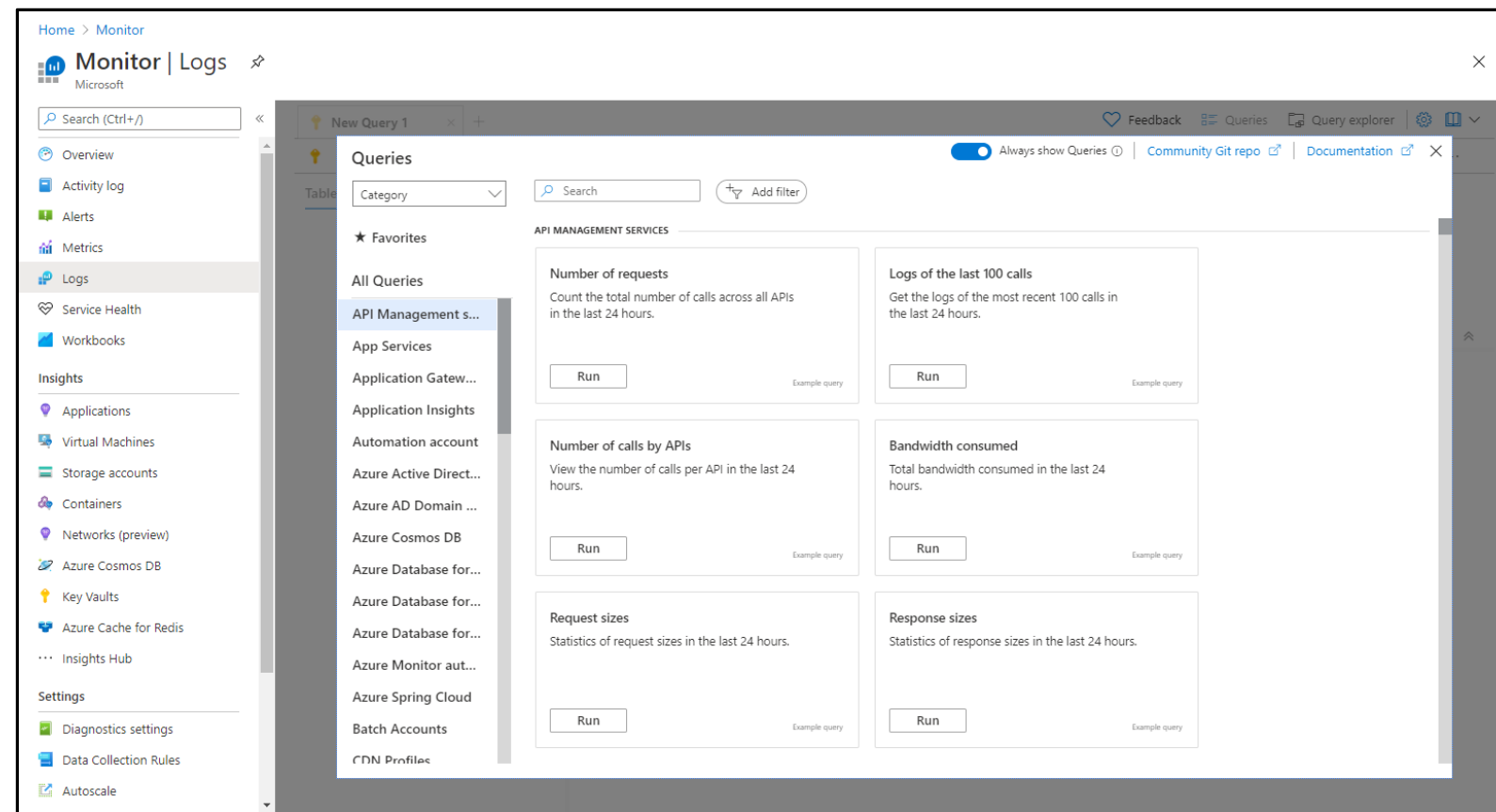
1. Sign in to the Azure portal
2. Go to the monitor page
3. Select alerts
4. Create new alert rules
5. Add the condition



Determine Log Analytics Uses

Log analytics is a service that helps the user:

- Collect and analyze data generated by resources in their cloud and on-premises environments
- Write log queries and interactively analyze their results
- Examples of this include assessing system updates and troubleshooting operational incidents



Source: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/log-analytics-overview>

Create a Workspace

Home > Log Analytics workspaces >

Create Log Analytics workspace

Basics Pricing tier Tags Review + Create

i A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#) ×

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Resource group * ⓘ [Create new](#)

Instance details

Name * ⓘ ✓

Region * ⓘ

[Review + Create](#) [« Previous](#) [Next : Pricing tier >](#)

Features of log analytics workspace:

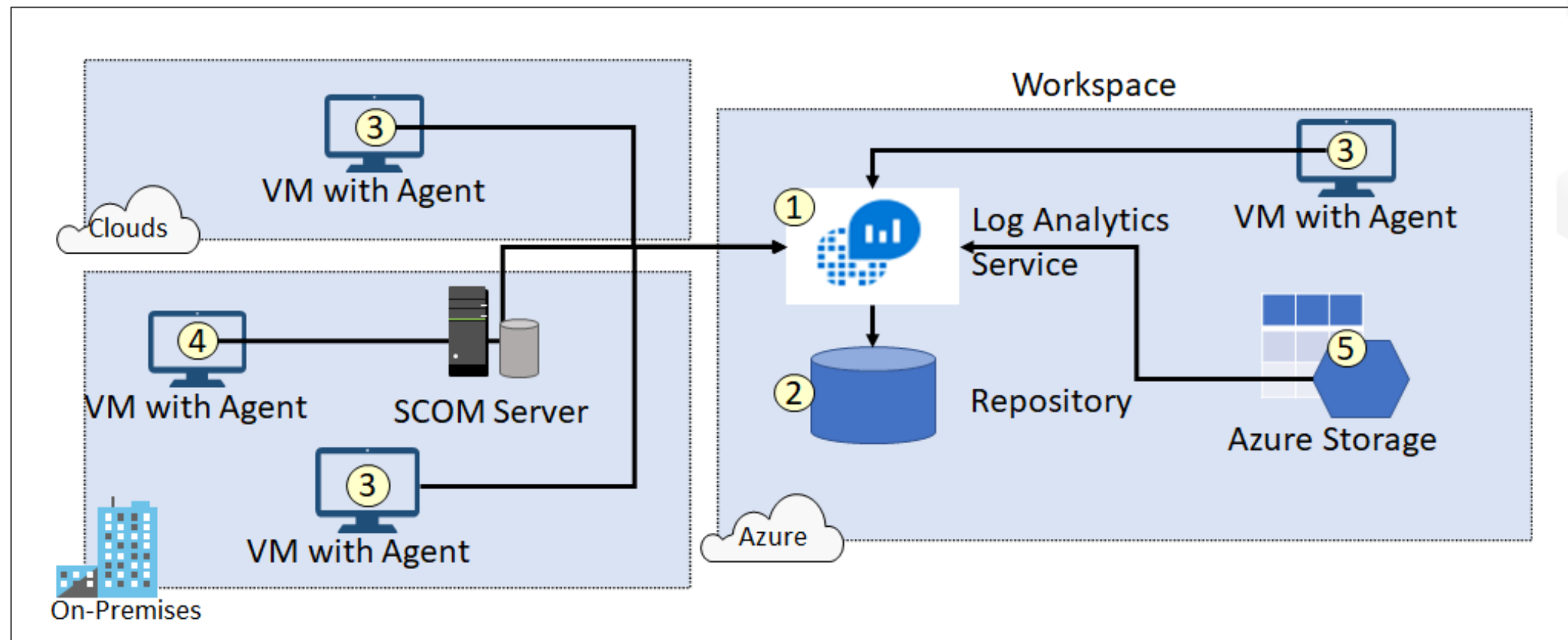
- A workspace is an Azure resource and a container where data is collected, aggregated, analyzed, and presented.
- The user can have multiple workspaces per Azure subscription, and they can have access to more than one workspace.
- A workspace provides a geographic location, data isolation, and scope.

Source: <https://learn.microsoft.com/en-us/azure/azure-monitor/logs/quick-create-workspace?tabs=azure-portal>

Create Connected Sources

Points to know about connected sources:

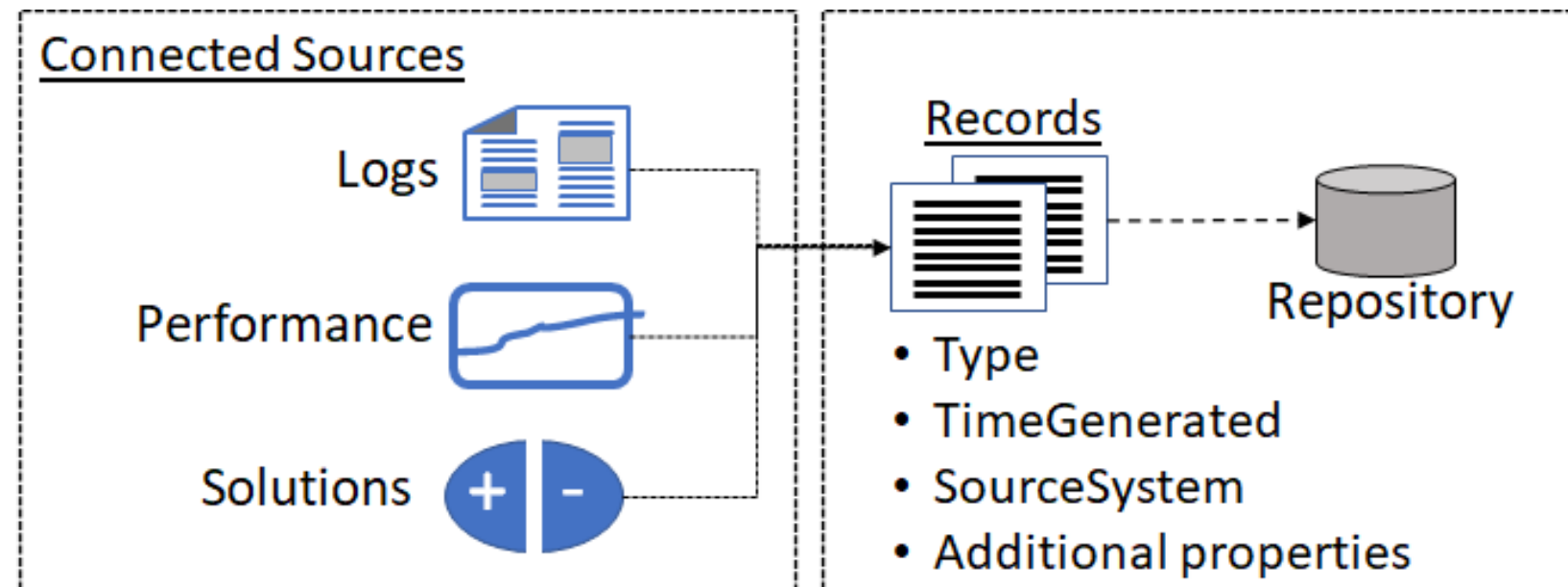
- Connected sources generate data.
- Data can be collected from Windows, Linux, SCOM, and Azure storage.



Define Data Sources

Data sources are the assorted data collected from each connected source. It can include events and performance data from Windows and Linux agents.

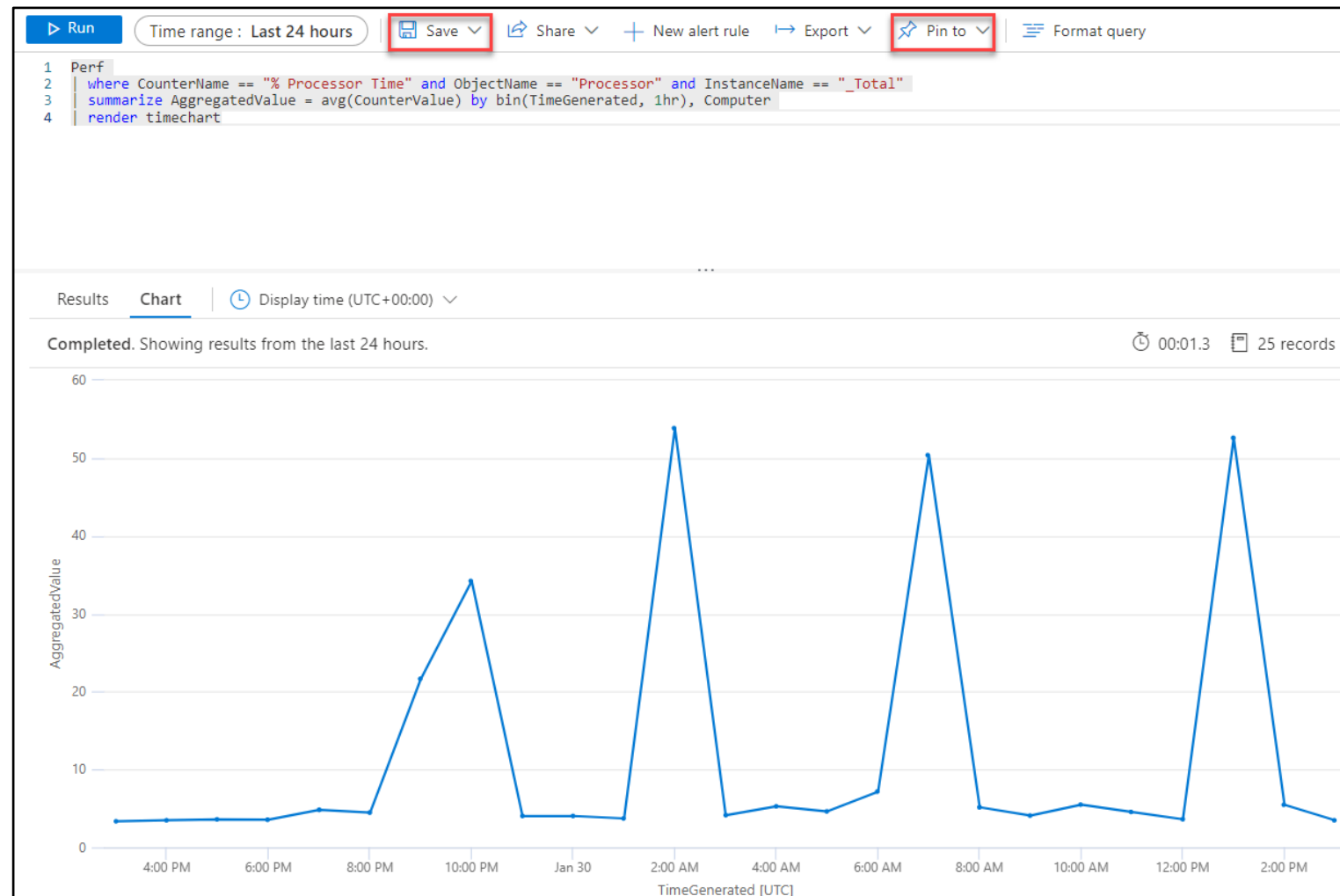
- Data sources include Windows event logs, Windows performance counters, Linux performance counters, IIS logs, custom fields, custom logs, and syslog.
- Each data source has additional configuration options.



Visualize Log Analytics Data

Provides a matter syntax to quickly retrieve and consolidate data within the repository

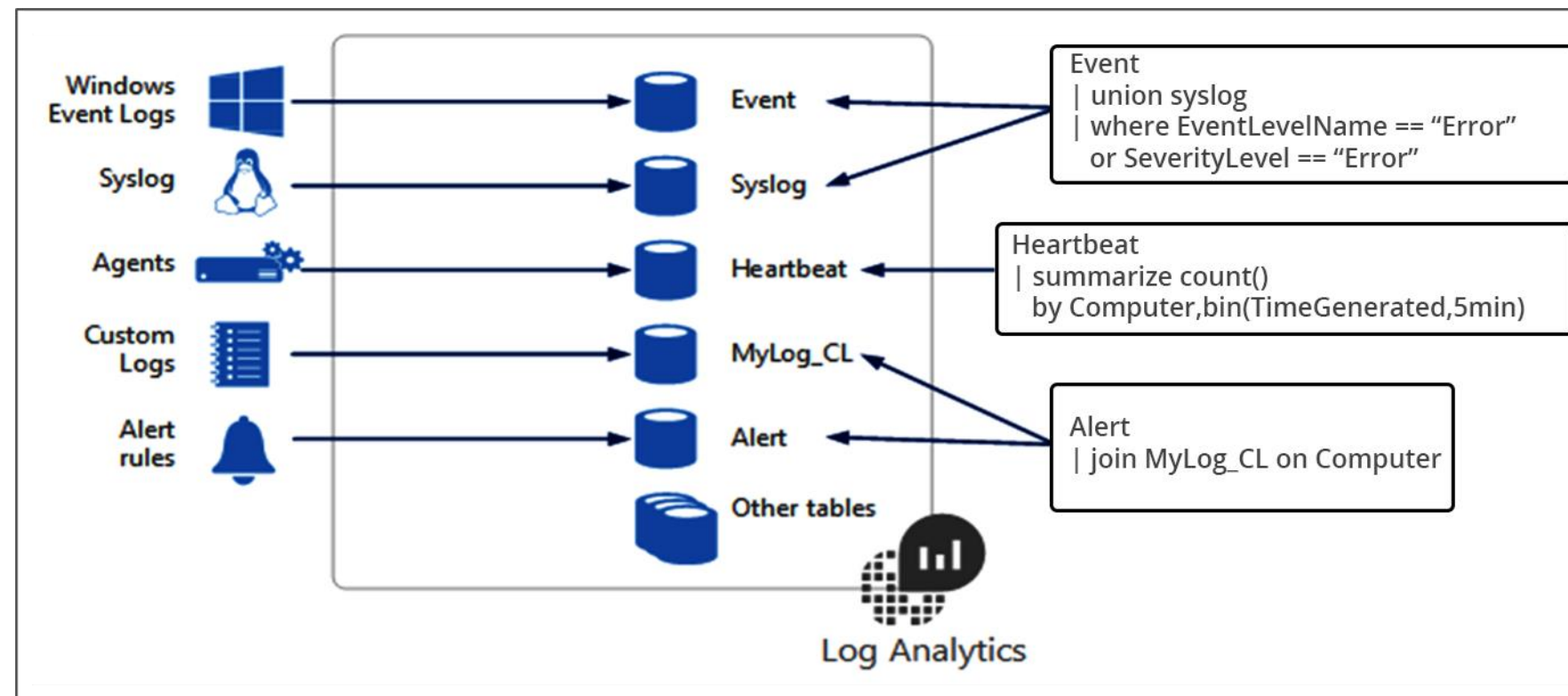
The features include:



- Provides a query syntax
- Retrieve and consolidate data in the repository quickly
- Save or have log searches run automatically to create an alert
- Export the data to Power BI or Excel

Structure Log Analytics Queries

When users build a query, they start by determining which tables have the required data. Each data source and solution stores its data in dedicated tables in the log analytics workspace.



```
Event
| where (EventLevelName ==
"Error")
| where (TimeGenerated >
ago(1 days))
| summarize ErrorCount =
count() by Computer
| top 10 by ErrorCount desc
```

Set Up Log Analytics Workspace



Duration: 10 Min.

Problem Statement:

You've been tasked to set up a log analytics workspace in the Azure portal and connect virtual machines to the workspace to collect and monitor VM logs

ASSISTED PRACTICE

Assisted Practice: Guidelines

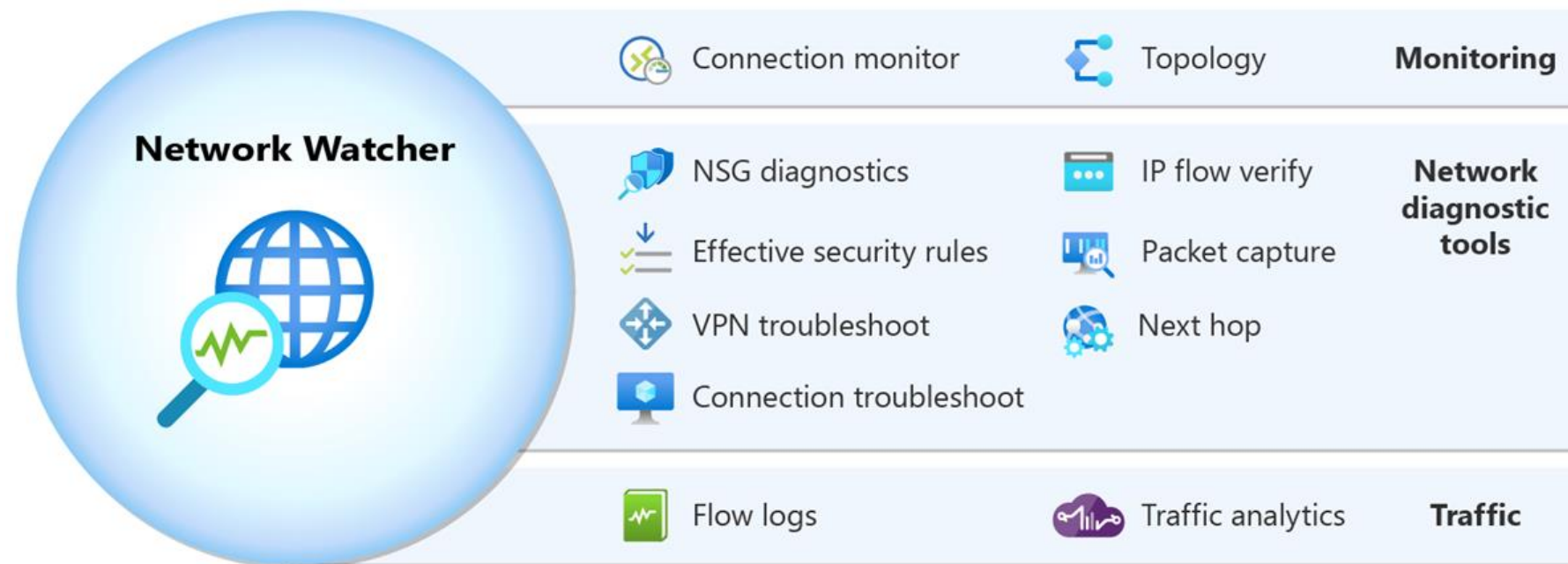
Steps to be followed:

1. Sign in to the Azure portal
2. Go to the Log Analytics page
3. Click Add
4. Provide the necessary details to create the workspace
5. Connect VMs to the workspace



Network Watcher

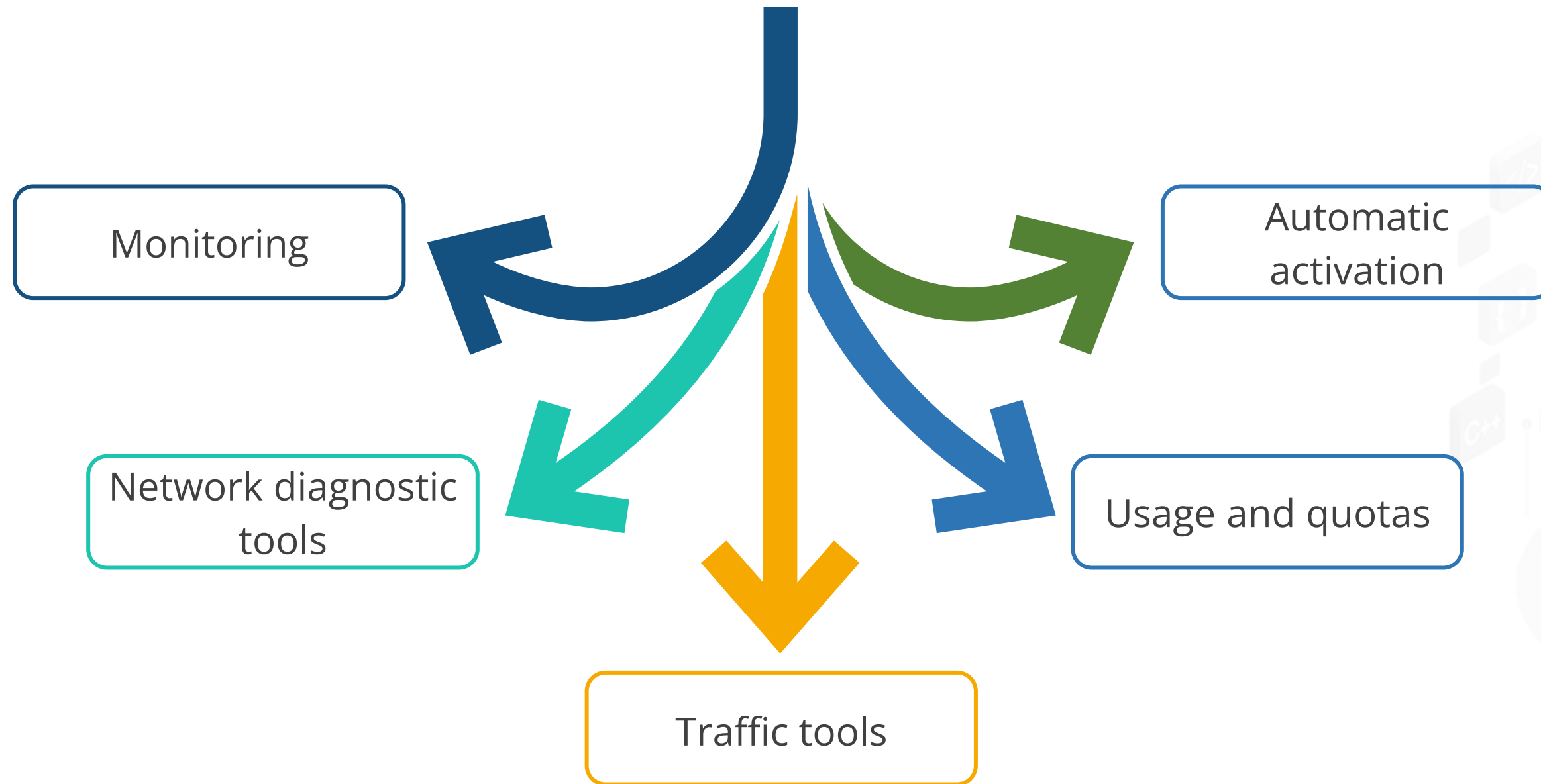
Network Watcher provides tools to monitor, diagnose, view metrics, and enable or disable logs for resources in an Azure virtual network.



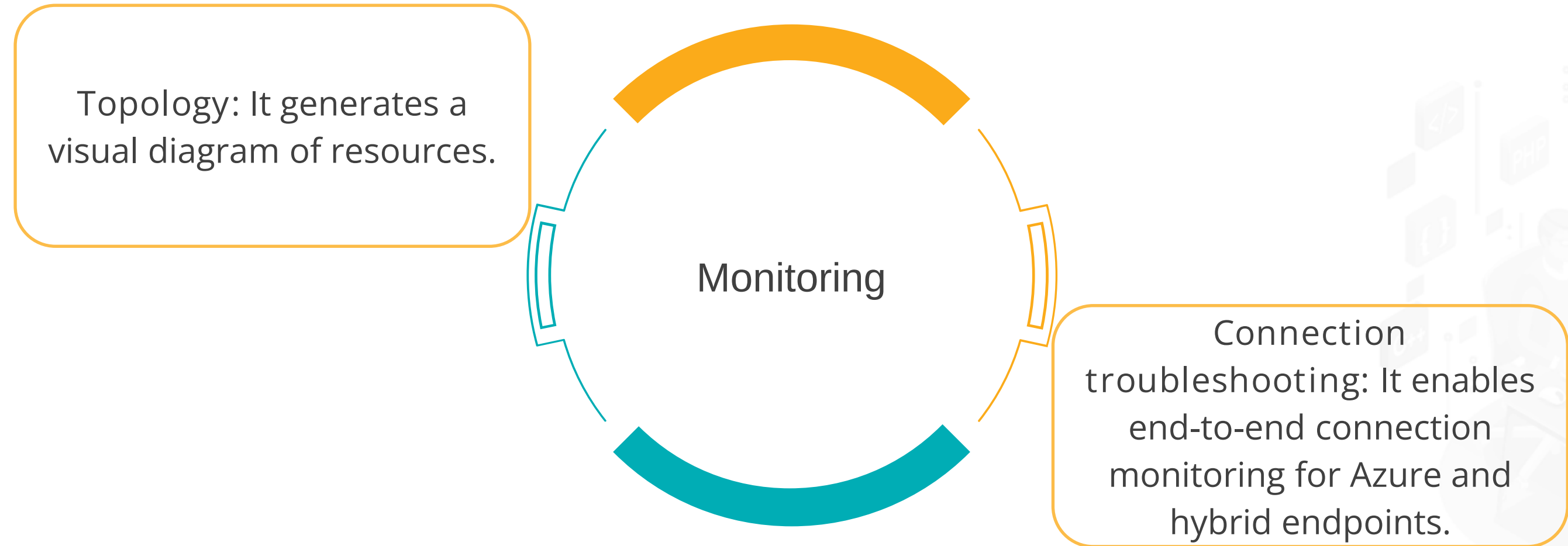
- Monitoring
- Network diagnostic tools
- Traffic

Network Watcher

Components and key features are:

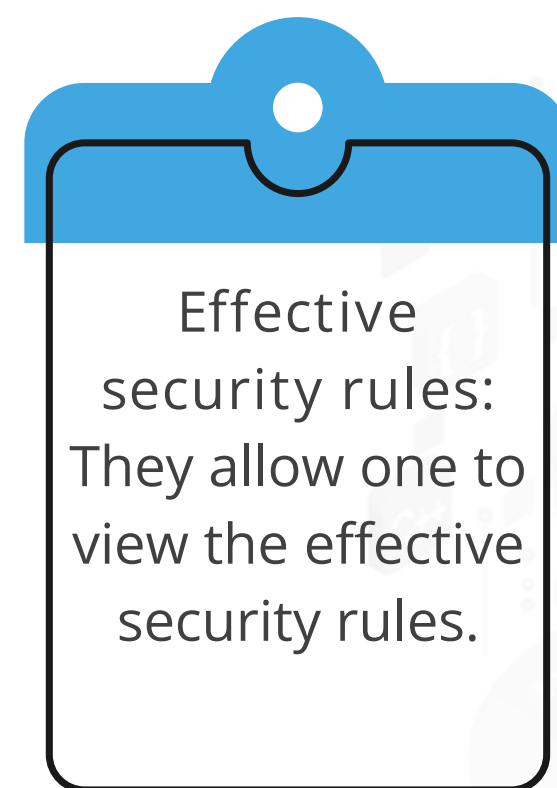
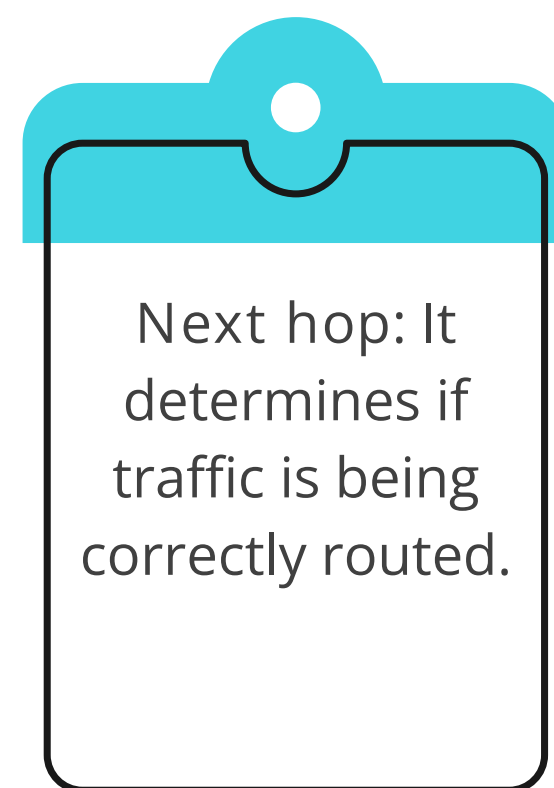
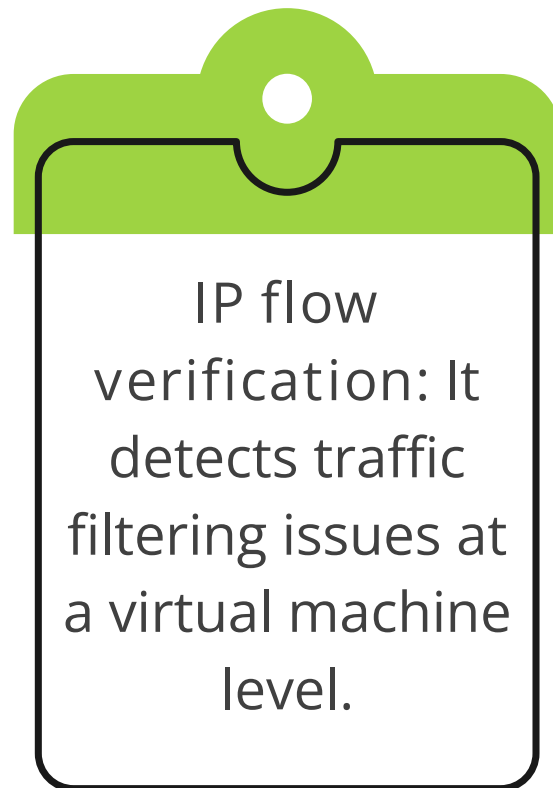


Network Watcher



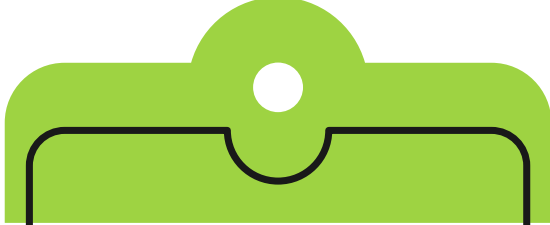
Network Watcher

Network diagnostic tools



Network Watcher

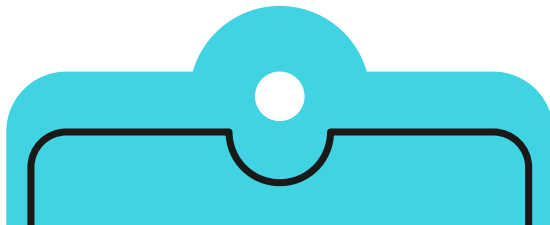
Network diagnostic tools



Connection troubleshooting
: It shows connections between various resources.



Packet capture:
It facilitates the remote creation of packet capture sessions.



VPN diagnostics:
They troubleshoot gateways and connections.



Network Watcher



Network Watcher

Usage and quotas

They offer the capability to summarize the usage of network resources in the Azure subscription and region.

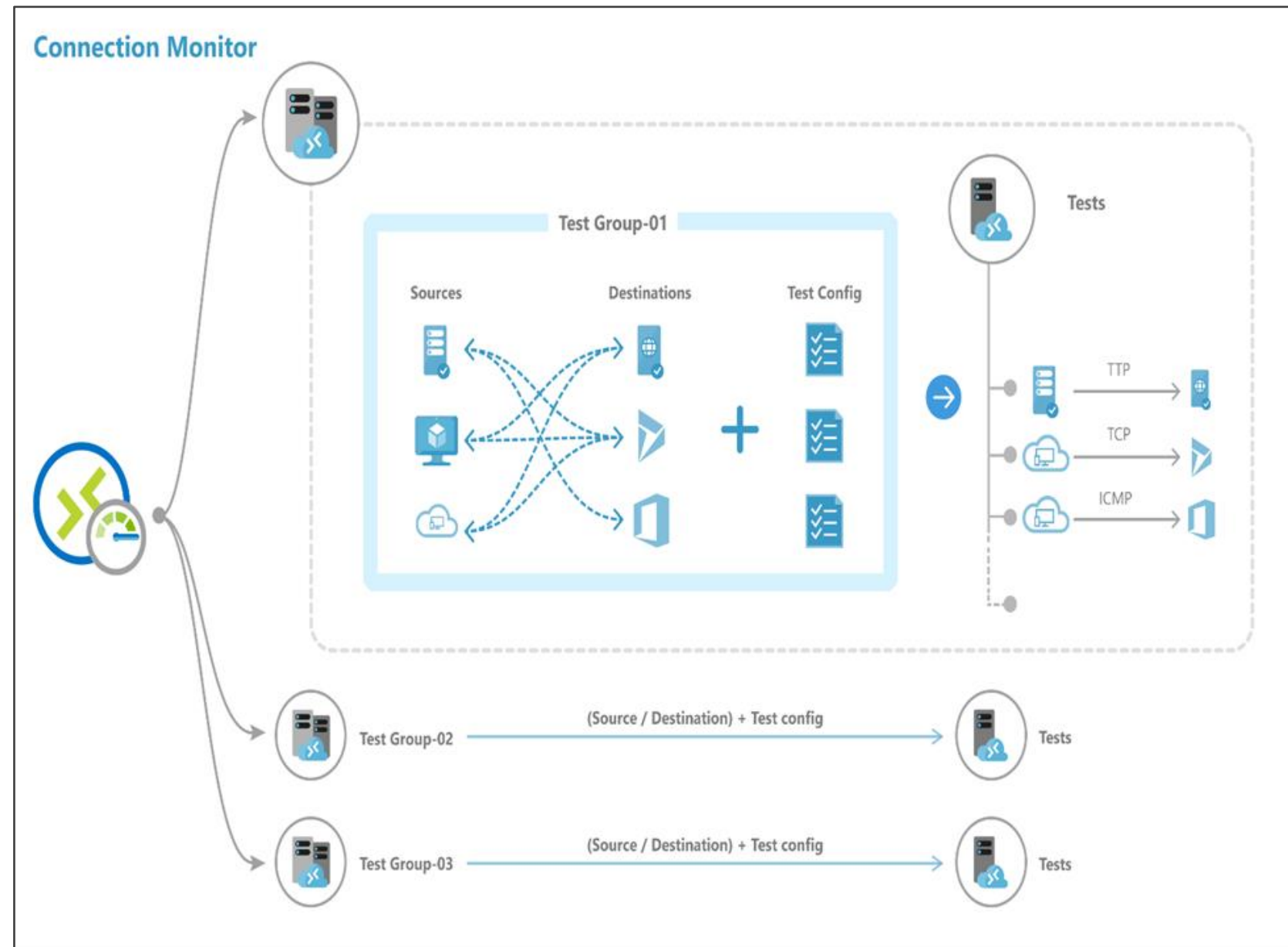
Automatic activation

It is automatically activated in the virtual network's region.



Configure Connection Monitor

It helps to monitor network connectivity between an Azure virtual machine and IP address.



Source: <https://learn.microsoft.com/en-us/azure/network-watcher/connection-monitor-overview>

Key Takeaways

- Azure Backup is the Azure-based service that users can use to back up (or protect) and restore their data in the Microsoft cloud.
- Recovery Services Vaults make it easy to organize their backup data while minimizing management overhead.
- Site Recovery helps ensure business continuity by keeping business apps and workloads running during outages.
- Azure Monitor enables the user to gather monitoring and diagnostic information about the health of their services.



Configure an Action Group in Case of Application Failure

Duration: 15 Min.



Project agenda: To implement an action group that will get the alerts in case of application failure

Description: A project to configure an action group so that the intended users get the alerts in case of any application failure

Perform the following:

Create action groups from Azure alerts and then configure the alerts based on application failure conditions.